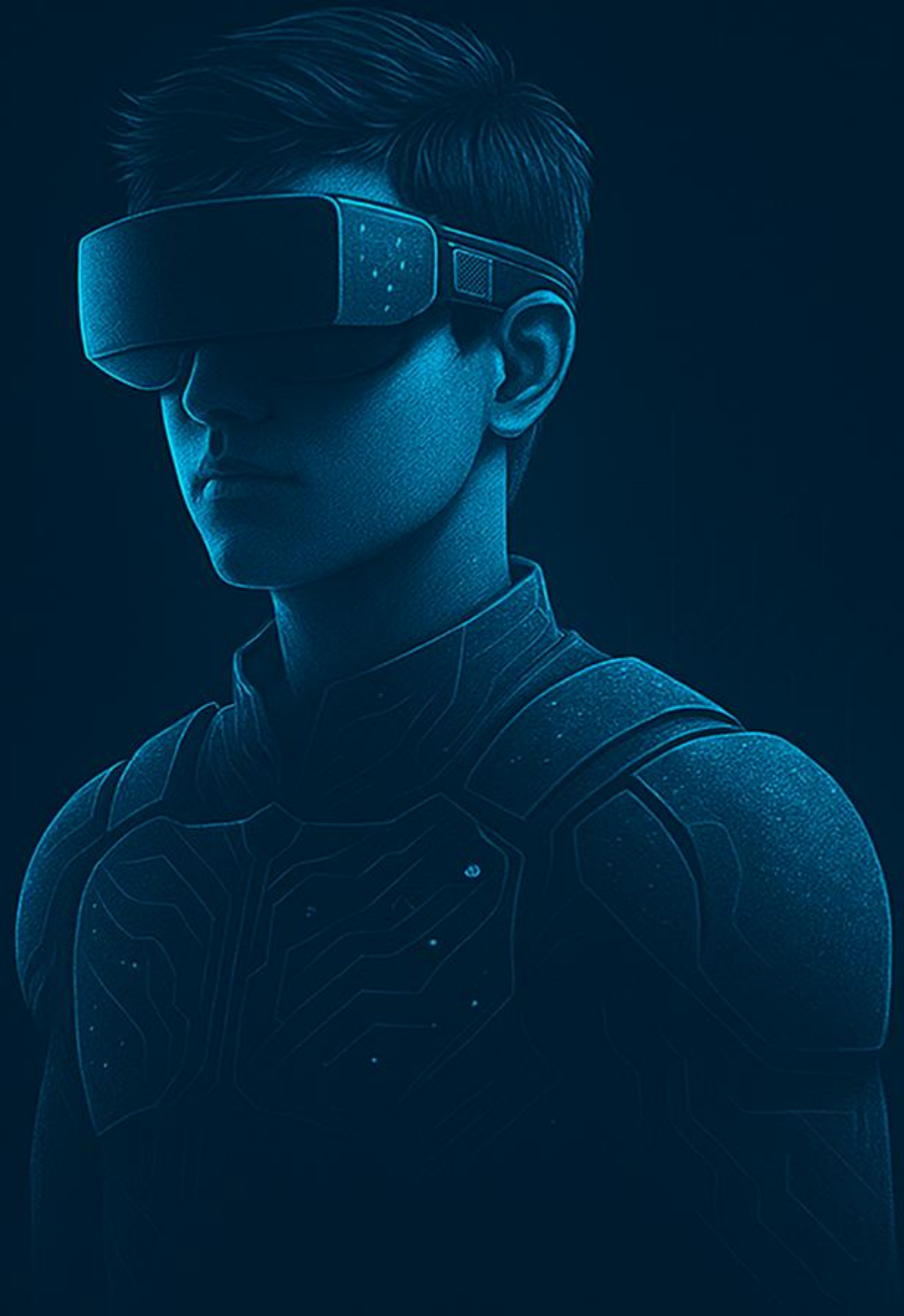


Module

1

Setting up the lab



Training structure



EASY



SIMULATION

Installation Labs



Agenda

- Duration: 6 hours (2 Theory + 4 Practical)
- Environment: Kali Linux (primary), Windows 10/11 (optional)
- Hypervisors: VirtualBox or VMware Workstation
- Level: Introductory - individual desktop labs

Goals

- Understand virtualization fundamentals and hypervisor concepts.
- Configure and manage personal virtual environments.
- Practice resource allocation, networking and snapshot management.
- Apply basic lab security principles (passwords, isolation, backups).

Theory Virtualization (2h)

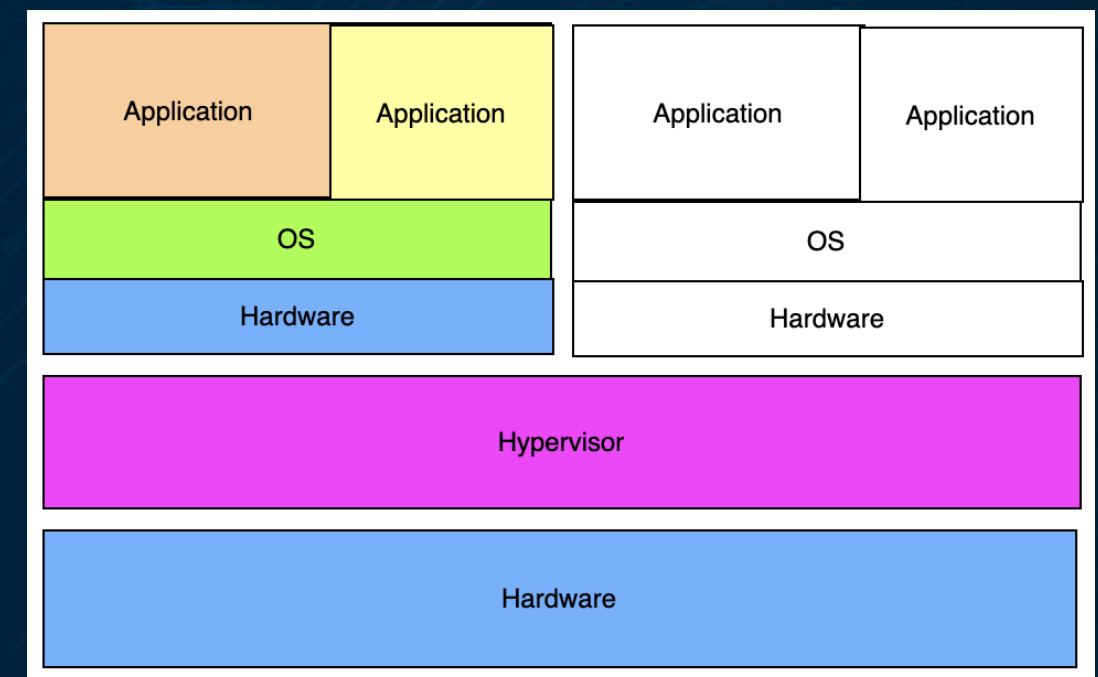
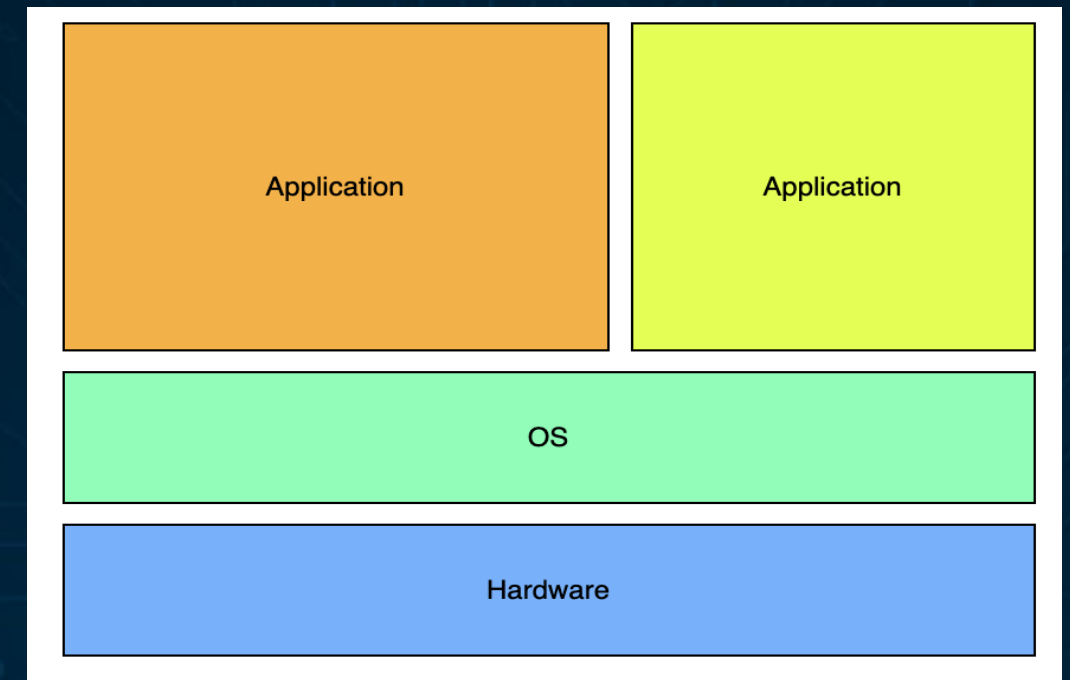


Objectives

- Define virtualization and its purpose in cybersecurity training.
- Explain differences between Type 1 and Type 2 hypervisors.
- Understand host vs guest resource allocation.

What is virtualization?

- One physical computer
- Multiple virtual machines
 - Different OS, but (often) shares architecture
- Each VM runs independently
 - Resources allocated per VM
 - Each has their own “interface”



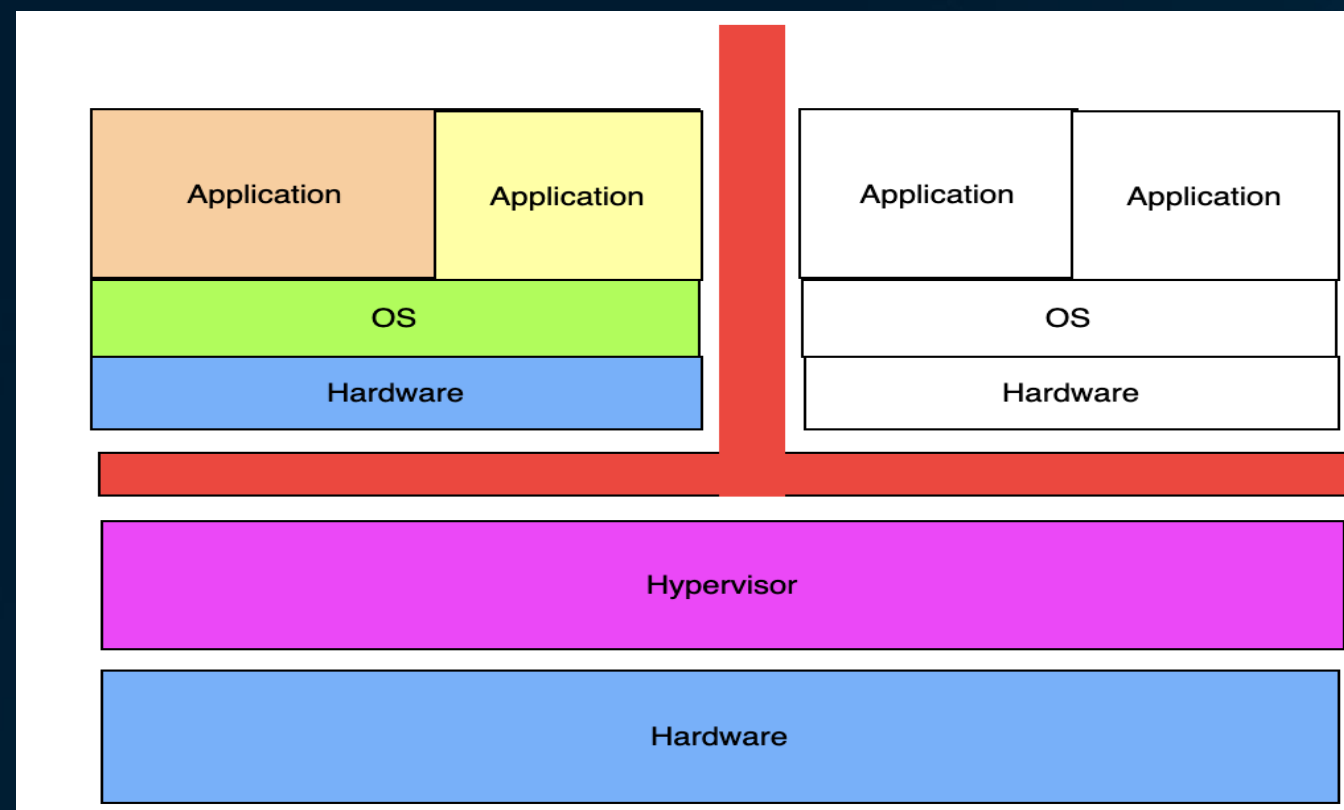
The Building analogy

- One building, many apartments
- Shared electricity and water
- Each apartment runs independently
- Different tenants (operating systems)



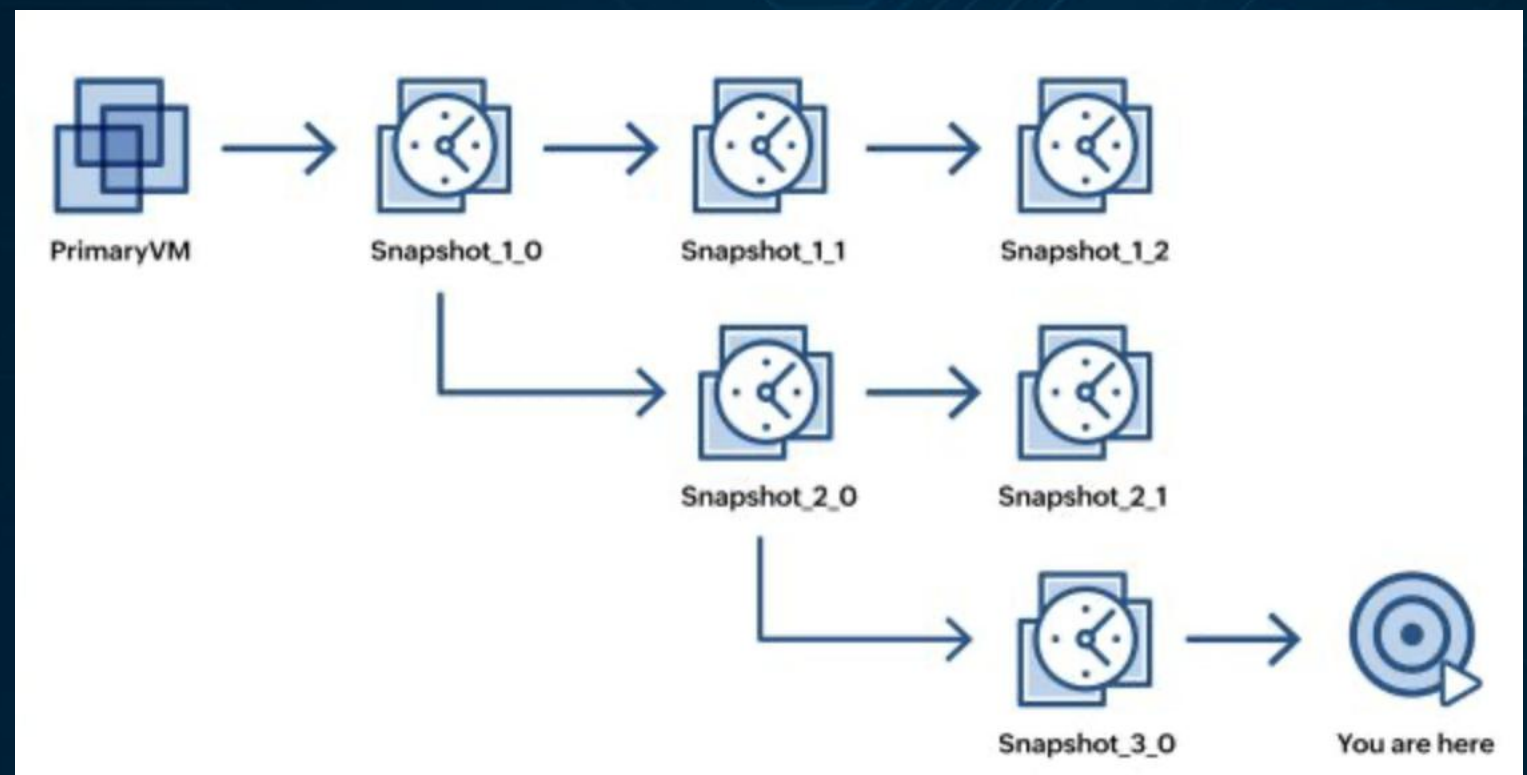
Why Use Virtualization?

- Safe testing environment
- Cost-effective (no extra hardware)
- Snapshot & rollback capability
- Isolation from host system



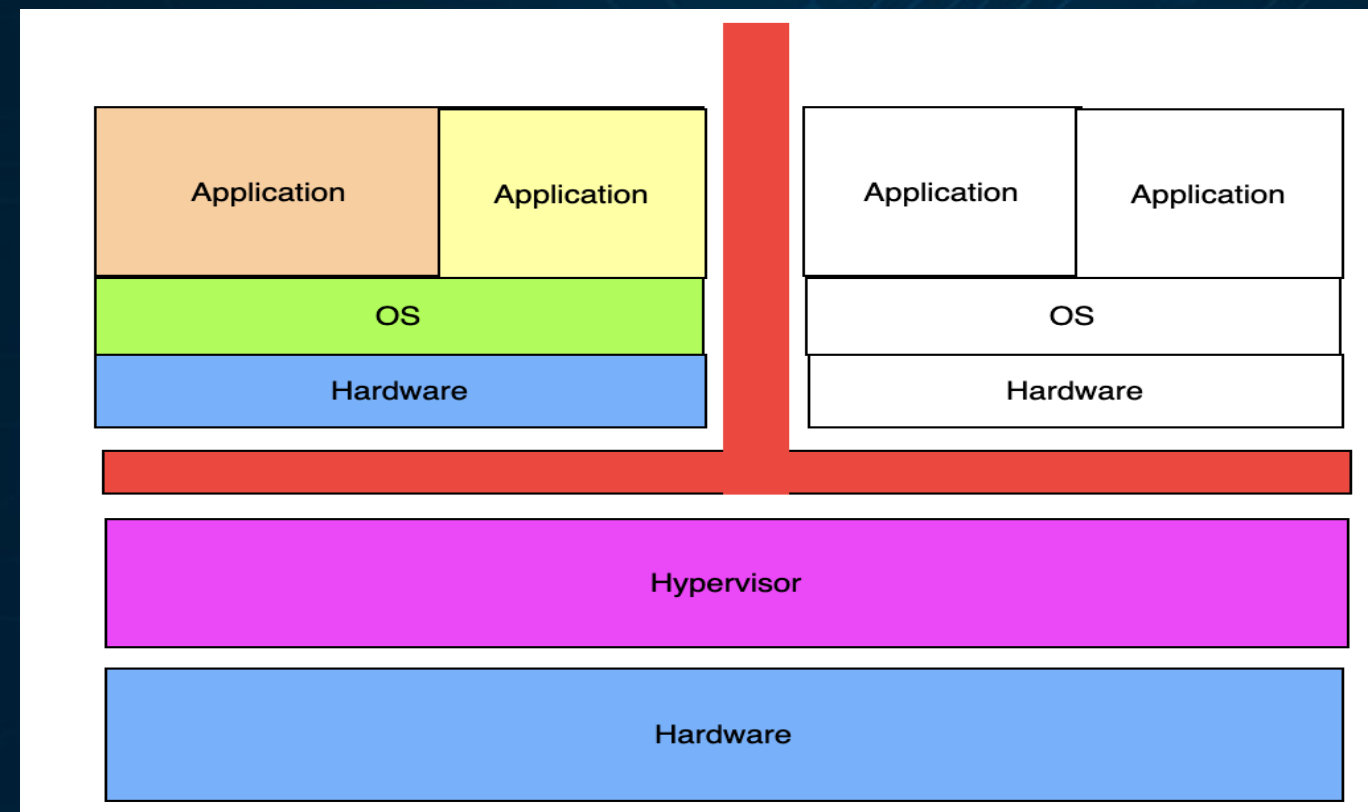
Snapshots = Time Machine

- Save current VM state
- Work safely, experiment freely
- Made mistake? Restore snapshot
- Multiple save points possible



Virtualization in Cybersecurity

- Simulate cyber attacks safely
- Build multi-machine networks
- Practice SOC monitoring
- Industry standard skill



Understanding Hypervisors

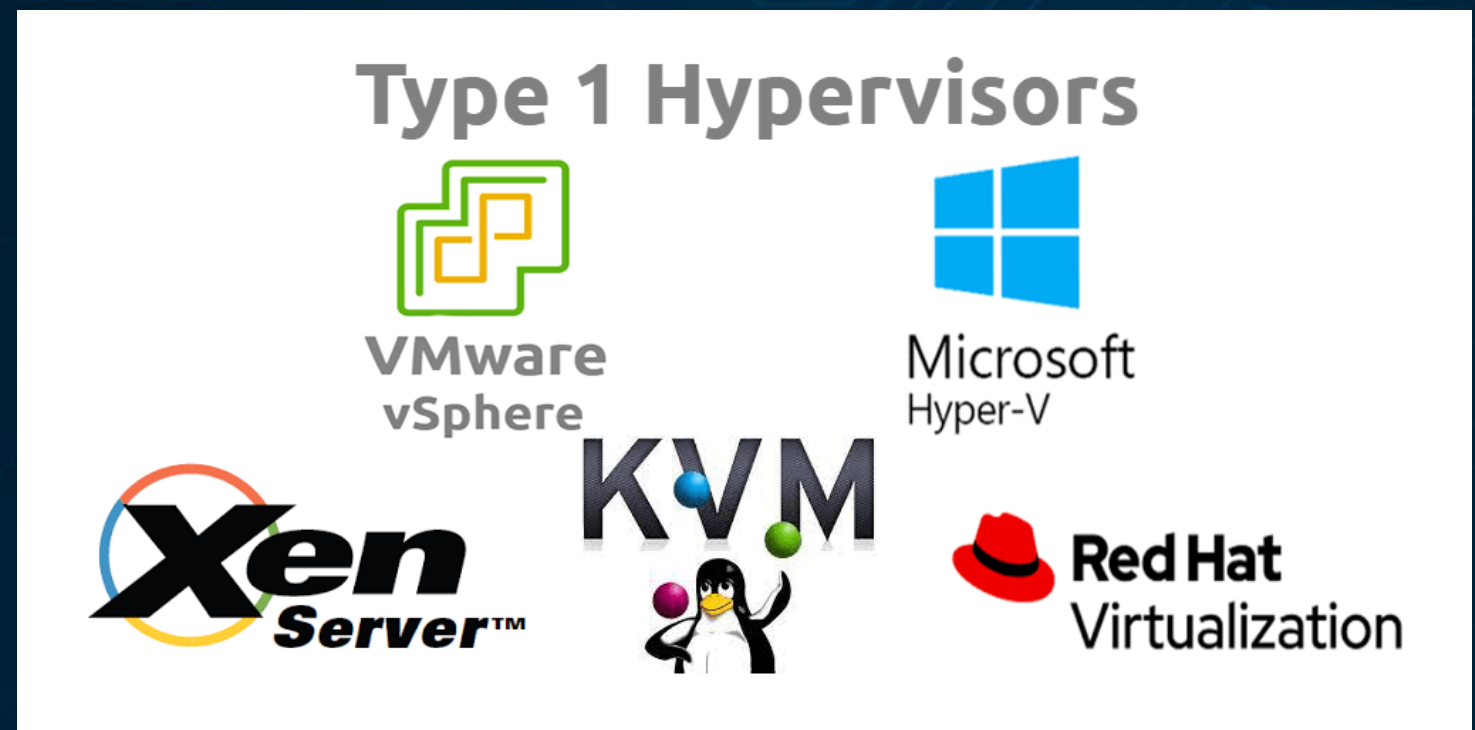
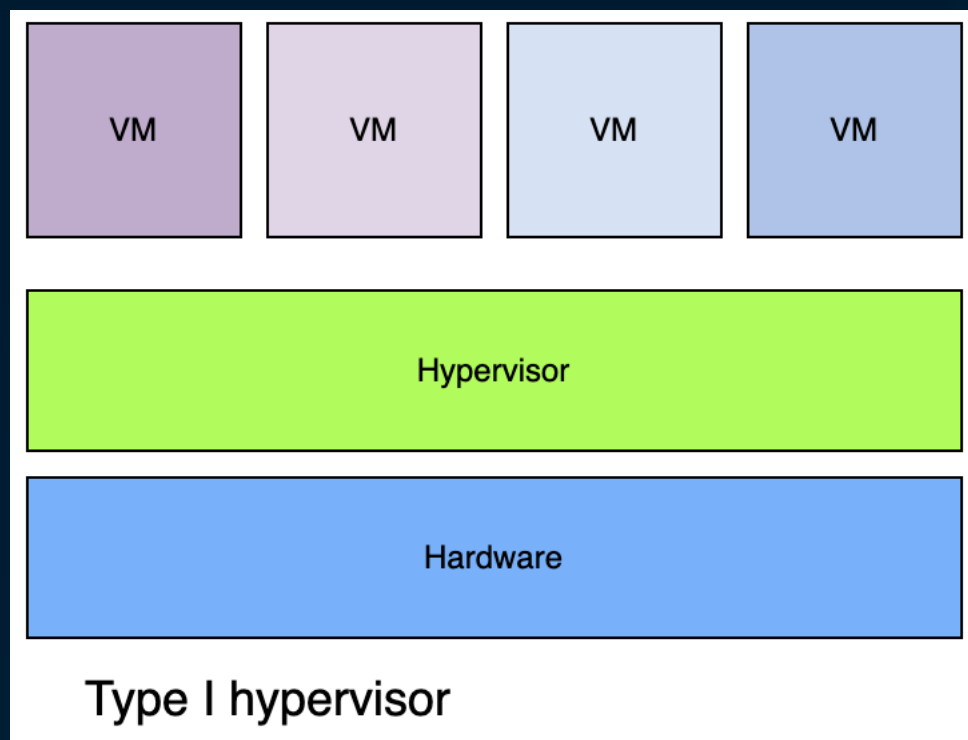
- What is a hypervisor
- Type 1 and 2 hypervisors
- What to pick when

What is a Hypervisor?

- Software managing virtual machines
- "Traffic controller" for resources
- Shares CPU, RAM, storage
- Also called: VMM (Virtual Machine Manager)

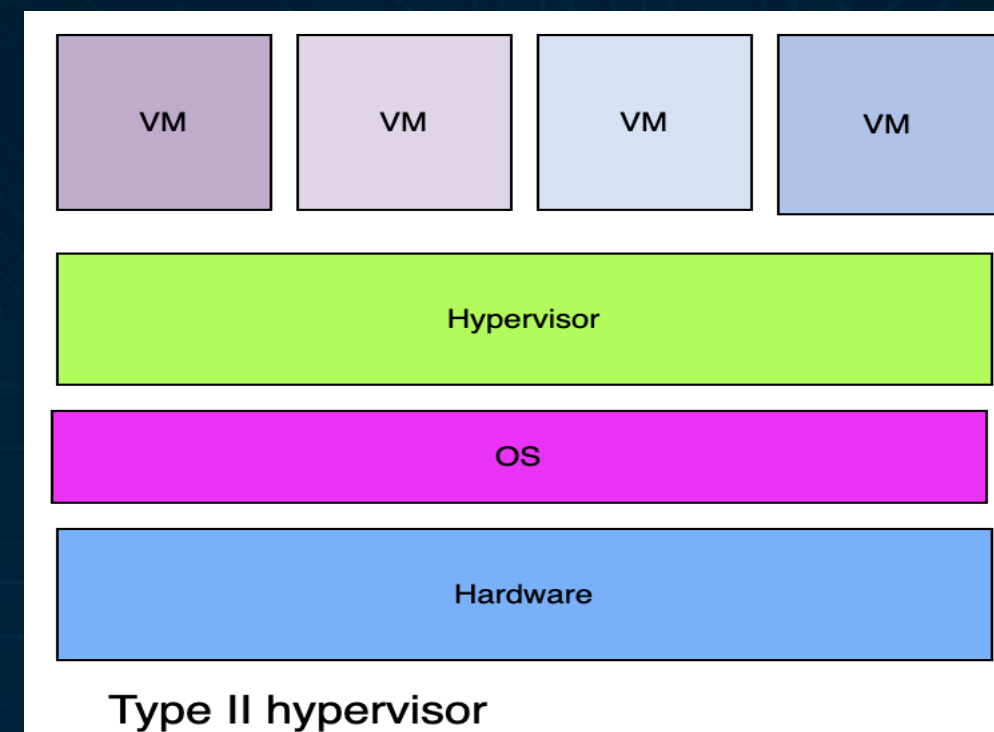
Type 1 Hypervisor (Bare Metal)

- Runs directly on hardware
- No operating system needed
- Fast and efficient
- Used in data centers



Type 2 Hypervisor (Hosted)

- Runs on top of OS
- Installed like normal application
- Easy to use
- Perfect for training



Comparison

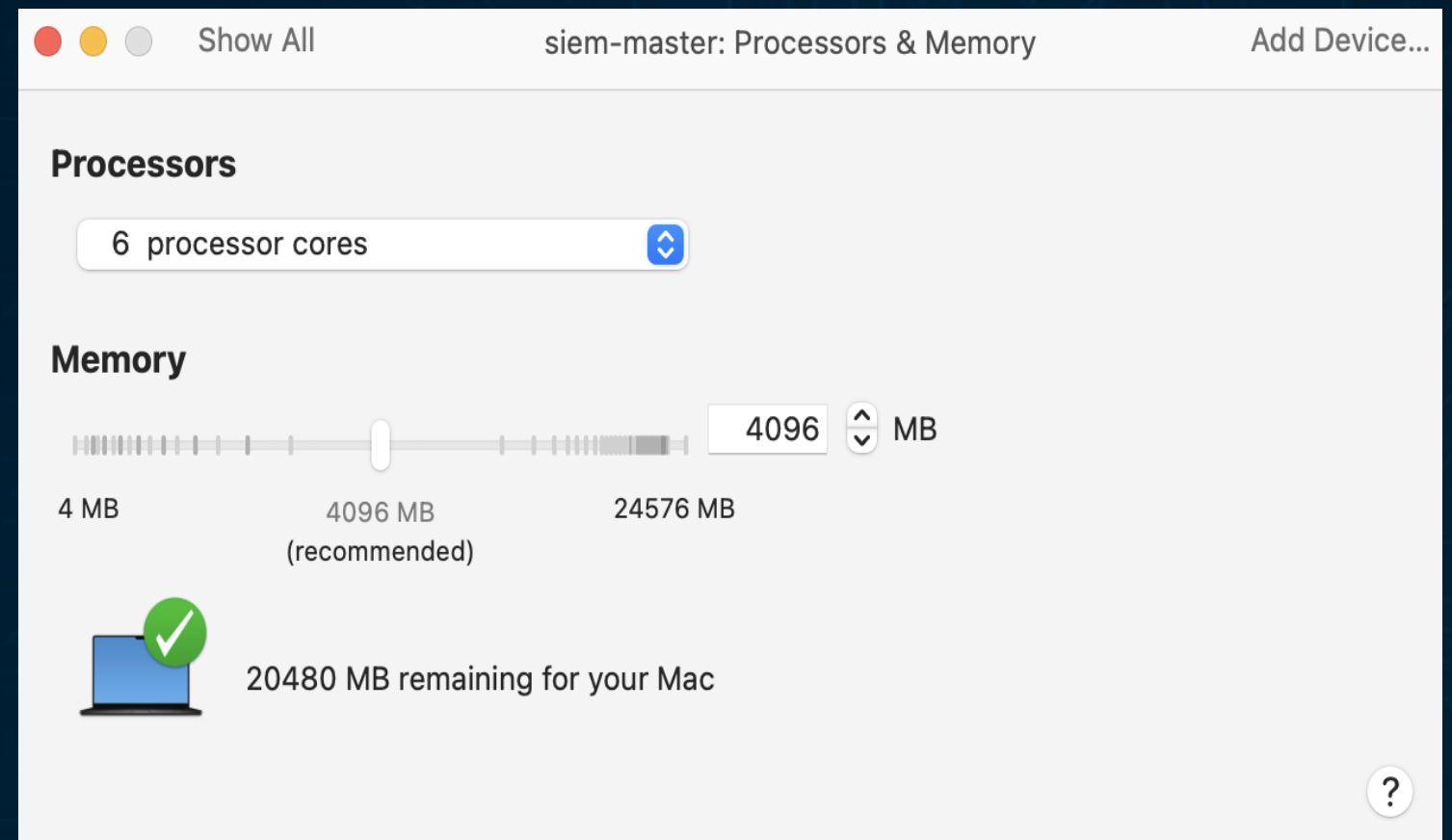
Type I	Type II
On hardware	On OS
Faster	Slower / good Speed
Espensive	Free Options Available
Data center/Servers	Home use/Laptops/Workstations

Our Choice: VirtualBox

- 100% FREE and open-source
- Works on Windows/Mac/Linux
- Easy to learn
- Great for cybersecurity training

Resource Allocation

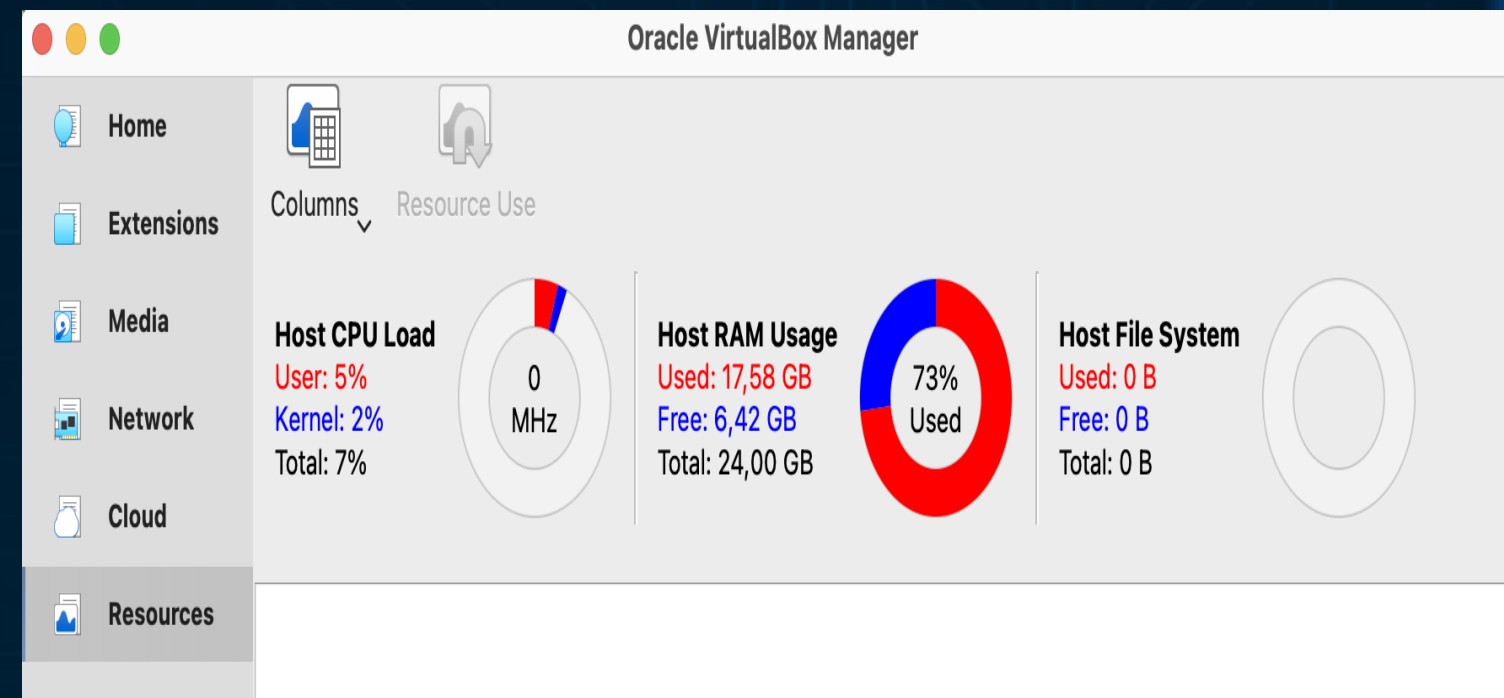
- Host vs Guest
- Sharing Resources
- Network Resources



Memory and CPU allocation in VMWare

Host and Guests

- **HOST** (Physical Computer)
 - Your real computer
 - Provides all resources
 - Runs the hypervisor
- **GUEST** (Virtual Machine)
 - Virtual computer
 - Borrows host resources
 - Can run different OS



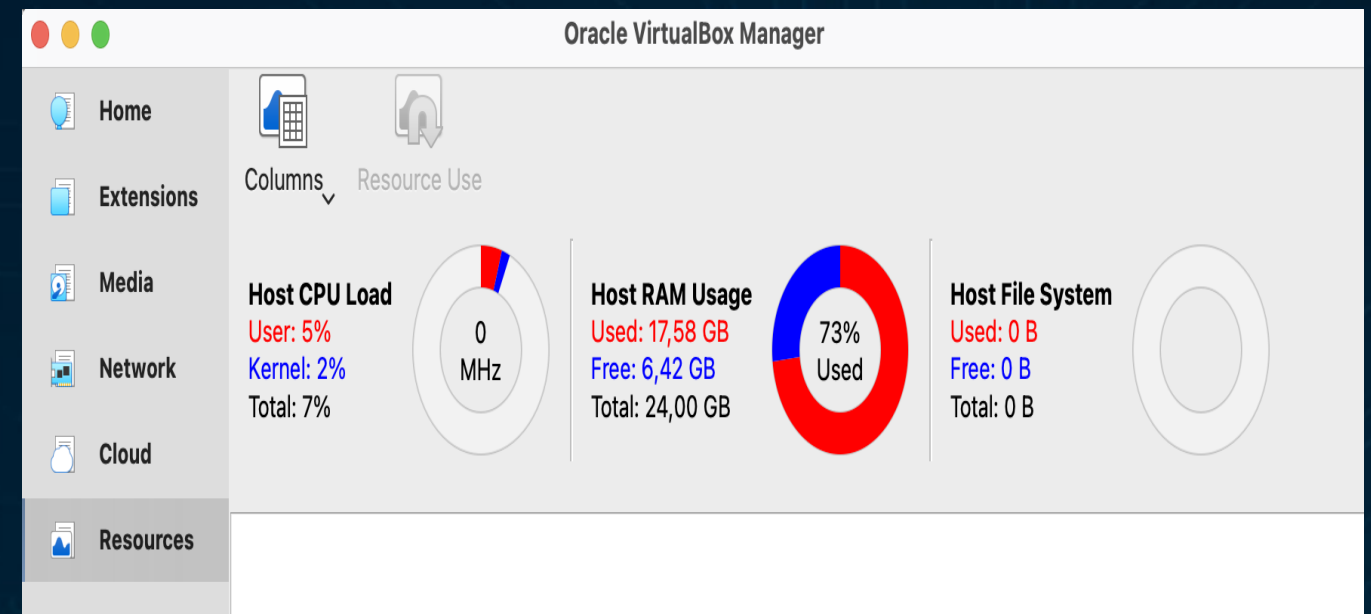
Resource usage on virtualbox without any guests

What Gets Shared?

-  CPU - Processing power
-  RAM - Memory for programs
-  Storage - Hard drive space
-  Network - Internet connectivity

The Balancing Act

- Total Resources
 - Host needs some
 - Hypervisor needs some
 - VMs share the rest



Resource usage on virtualbox without any guests

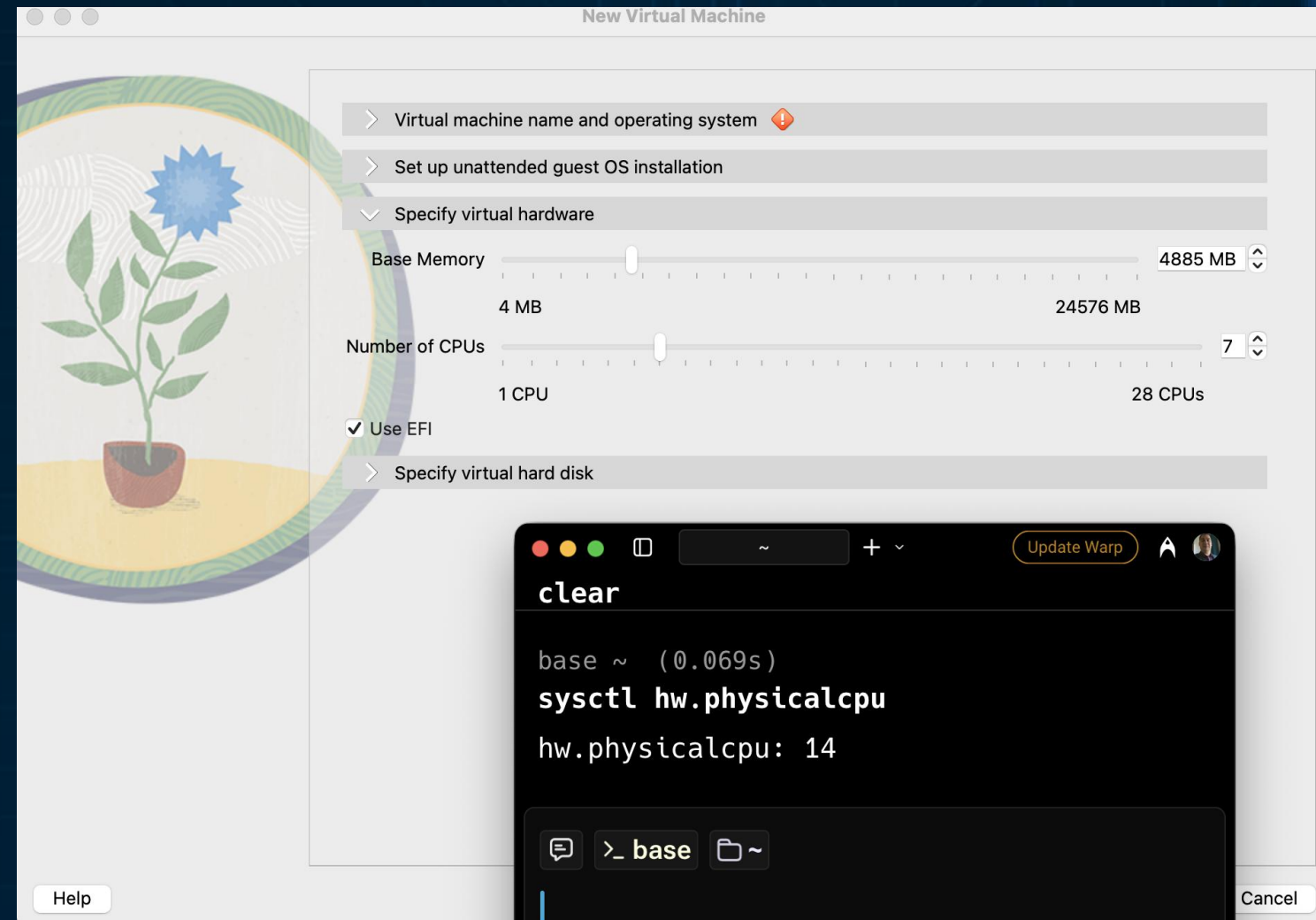
- Golden Rule: Can't allocate >80 % to VMs without resource starvation!

Example: 16GB System

- Your System
 - 16GB RAM total
 - Host needs: 4GB
 - Hypervisor: 2GB
- Available for VMs: 10GB

CPU Allocation Tips

- Don't exceed physical cores
- Reserve 1 core for host
- 1-2 cores per VM typical
- Over-subscription OK if not simultaneous



CPU allocation in virtualbox and physical count of CPU's
from the commandline in the Host system

Storage Allocation

- Static: Fixed size upfront
 - Better performance
 - Uses space immediately
- Dynamic: Grows as needed
 - Saves disk space
 - Slight performance cost

Snapshots = Time Machine

- Save current VM state
- Work safely, experiment freely
- Made mistake? Restore snapshot
- Multiple save points possible

How Snapshots Work

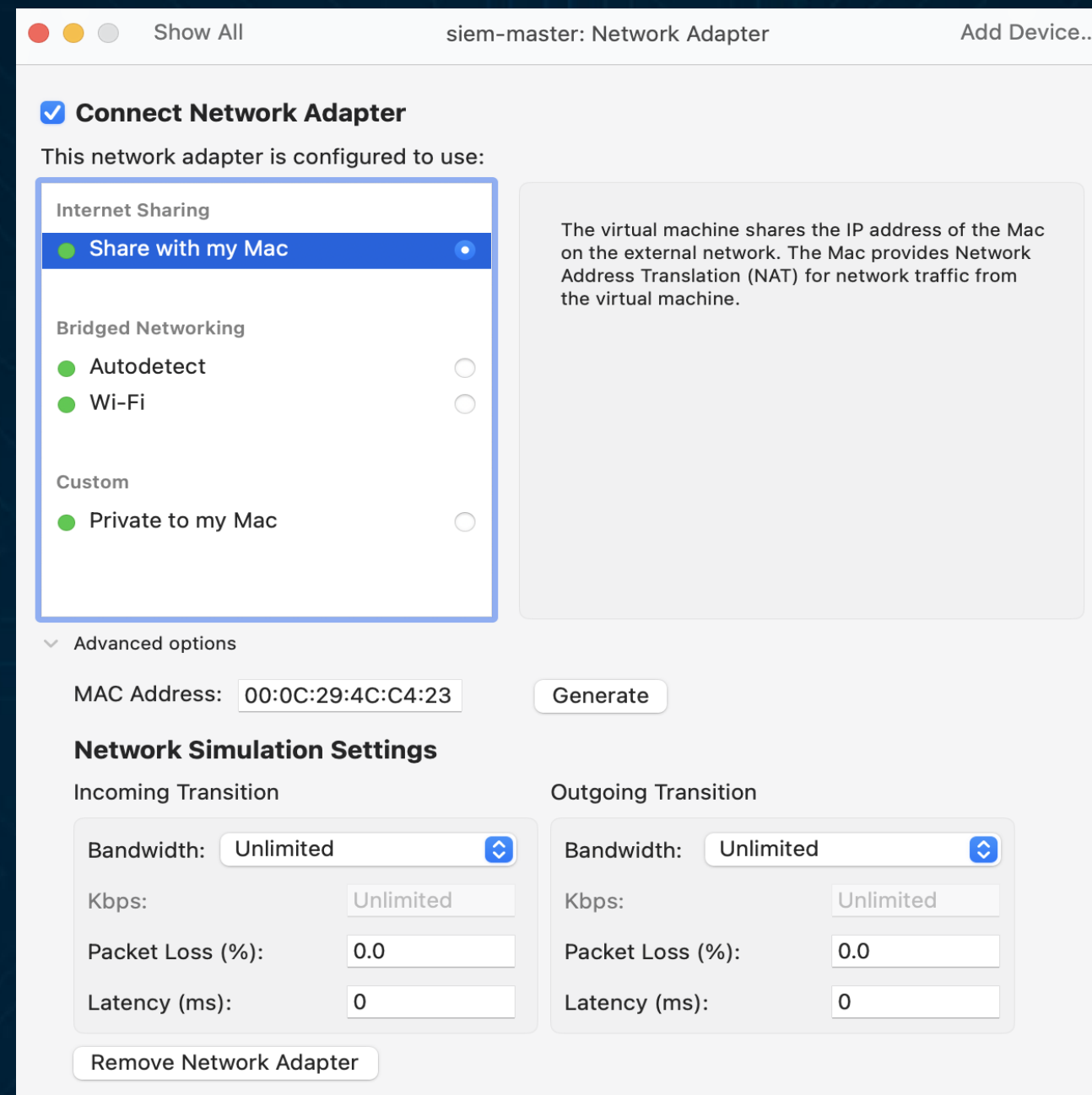
- Before Snapshot:
 - Clean system state
- During Work:
 - Install tools, test malware
 - System gets modified
- After Restore:
 - Back to clean state instantly!

Snapshot Best Practices

- Snapshot before risky changes
- Name snapshots clearly
- Don't keep too many
- Snapshots use disk space!

VM Network Types Overview

- Four main network modes:
- NAT
- Bridged
- Host-Only
- Internal



The screenshot shows the 'siem-master: Network Adapter' settings window in VMware. The 'Connect Network Adapter' checkbox is checked. Under 'Internet Sharing', 'Share with my Mac' is selected. Under 'Bridged Networking', 'Autodetect' and 'Wi-Fi' are available. Under 'Custom', 'Private to my Mac' is available. A description on the right states: 'The virtual machine shares the IP address of the Mac on the external network. The Mac provides Network Address Translation (NAT) for network traffic from the virtual machine.' Below, 'Advanced options' are expanded, showing the MAC Address '00:0C:29:4C:C4:23' and a 'Generate' button. 'Network Simulation Settings' are also shown for both Incoming and Outgoing transitions, with Bandwidth set to 'Unlimited', Kbps set to 'Unlimited', Packet Loss (%) set to '0.0', and Latency (ms) set to '0'. A 'Remove Network Adapter' button is at the bottom.

Network setting vmware on MacOS

NAT Network Mode

- Network Address Translation
 - VM shares host's IP
 - Can access internet 
 - Hidden from external network
 - Default network mode
- Best for: General use, safe browsing

Bridged Network Mode

- Direct Network Connection
 - VM gets own IP address
 - Appears as separate device
 - Full network access
- Best for: Network testing, servers

Host-Only Network Mode

- Isolated Communication
 - VM talks ONLY to host
 - No internet access
 - Complete isolation ⚠
- Best for: Malware analysis, secure testing

Common Problems

VM runs slow? → Add more RAM

Host freezes? → Reduce VM resources

Out of disk space? → Delete old snapshots

Guests do not start or are extremely slow? → Check if CPU virtualization capabilities are turned on

Virtualization (4h)

Practical



Objectives & Prerequisites

- What You'll Build:
 - VirtualBox hypervisor installed and configured
 - Kali Linux VM (security testing platform)
 - Windows 11 VM (enterprise environment simulation)
 - Working snapshots for both VMs
 - Configured and tested network connectivity
- Prerequisites:
 - Laptop/PC with at least 8GB RAM (16GB recommended)
 - 100GB free disk space minimum
 - Admin rights on your computer
 - Stable internet connection

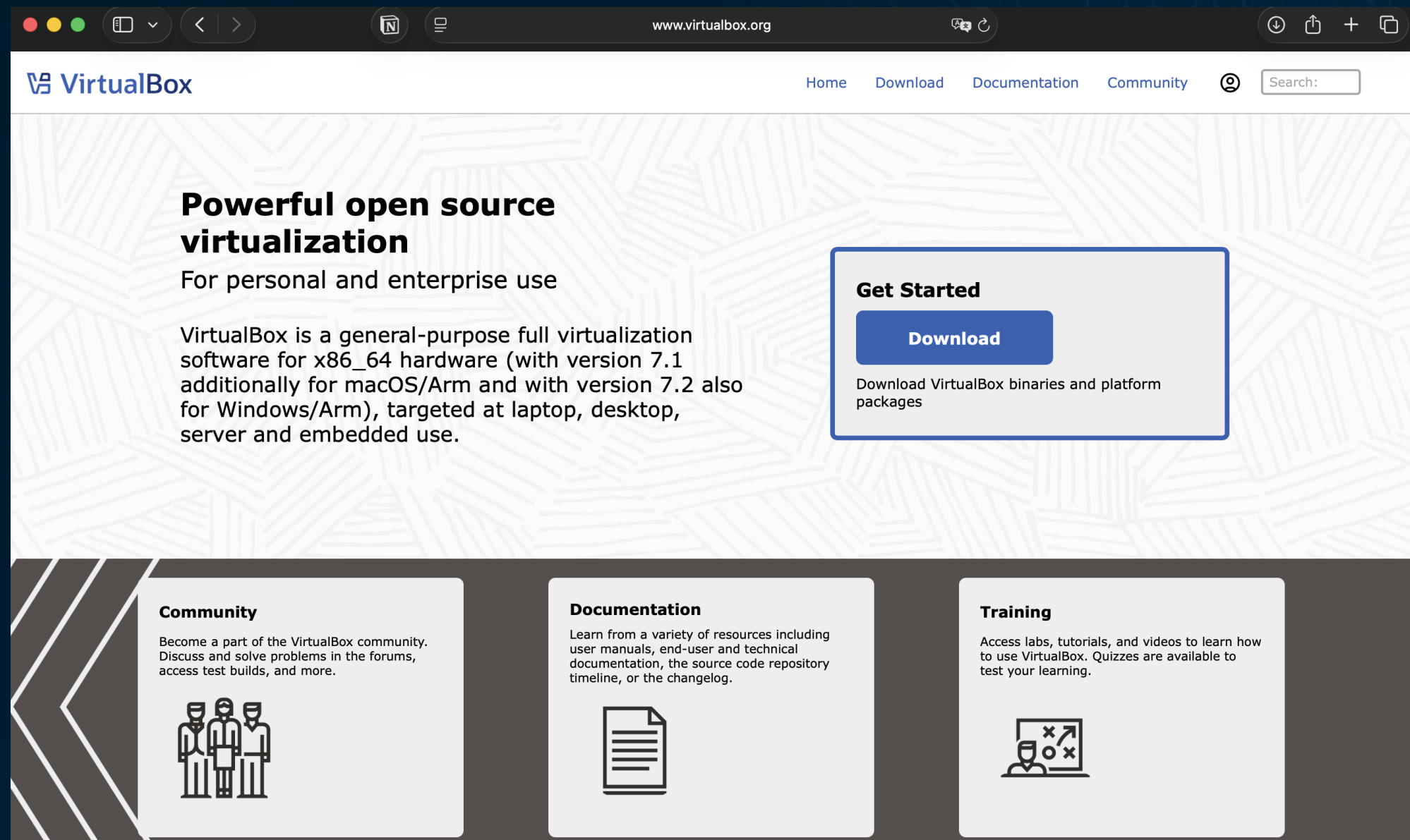
Lab 1: Install VirtualBox

Download Kali Linux

1. Open web browser
2. Navigate to: `https://www.virtualbox.org``
3. Click "Download VirtualBox"
4. Select your operating system:
 - Windows hosts
 - macOS hosts
 - Linux distributions
5. Download also: VirtualBox Extension Pack (adds USB 2.0/3.0 support)

Download Virtualbox

Install Virtualbox from www.virtualbox.org

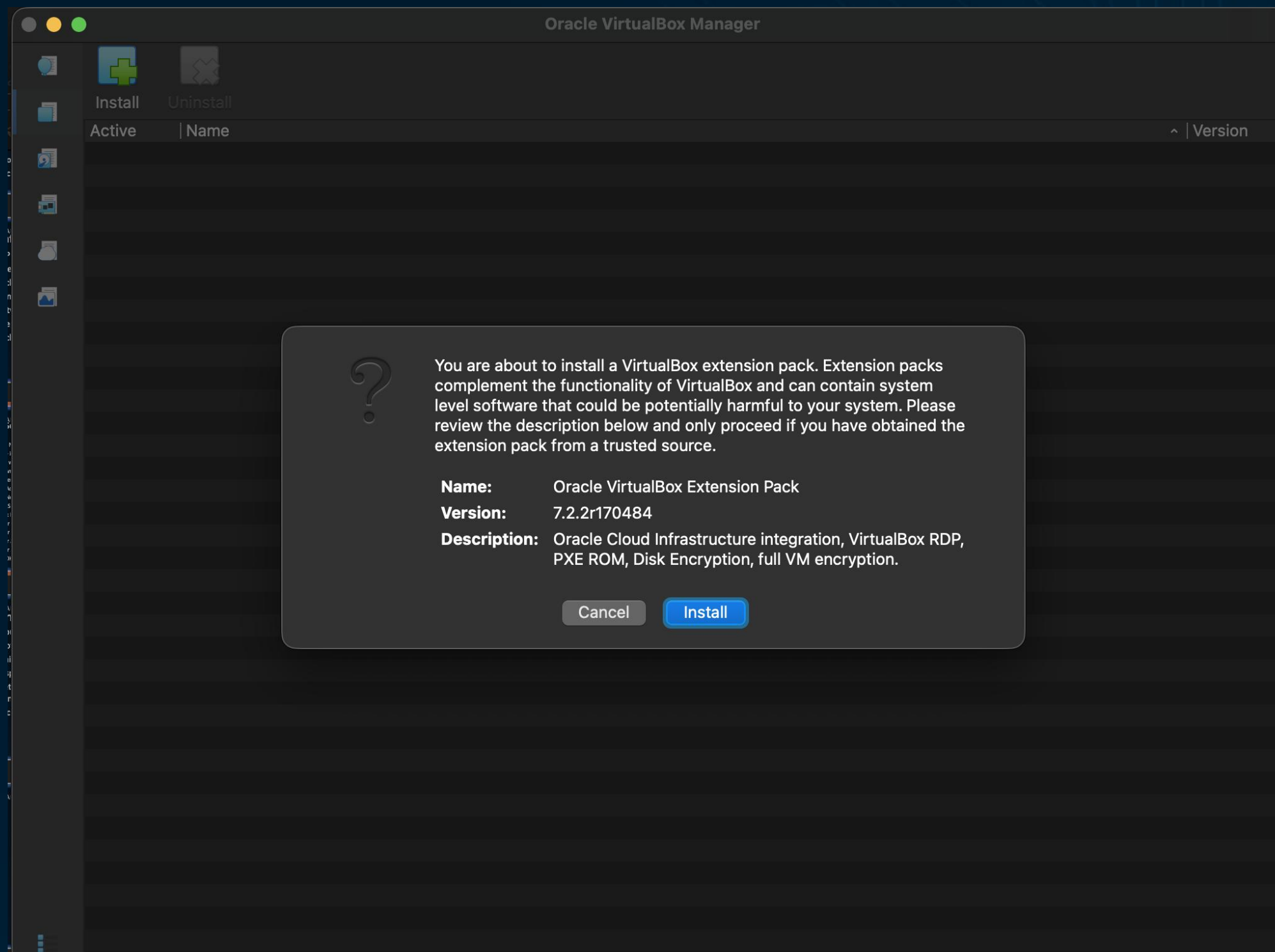


The screenshot shows the VirtualBox website in a web browser. The browser's address bar displays www.virtualbox.org. The website's navigation bar includes links for Home, Download, Documentation, and Community, along with a search bar. The main content area features a large heading "Powerful open source virtualization" and a subheading "For personal and enterprise use". Below this, a paragraph describes VirtualBox as a general-purpose full virtualization software for x86_64 hardware (with version 7.1) and additionally for macOS/Arm and with version 7.2 also for Windows/Arm, targeted at laptop, desktop, server and embedded use. To the right of this text is a "Get Started" box containing a prominent "Download" button and the text "Download VirtualBox binaries and platform packages". At the bottom of the page, there are three columns: "Community" (with an icon of three people and text about becoming part of the community), "Documentation" (with a document icon and text about learning from various resources), and "Training" (with an icon of a person at a computer and text about accessing labs, tutorials, and videos).

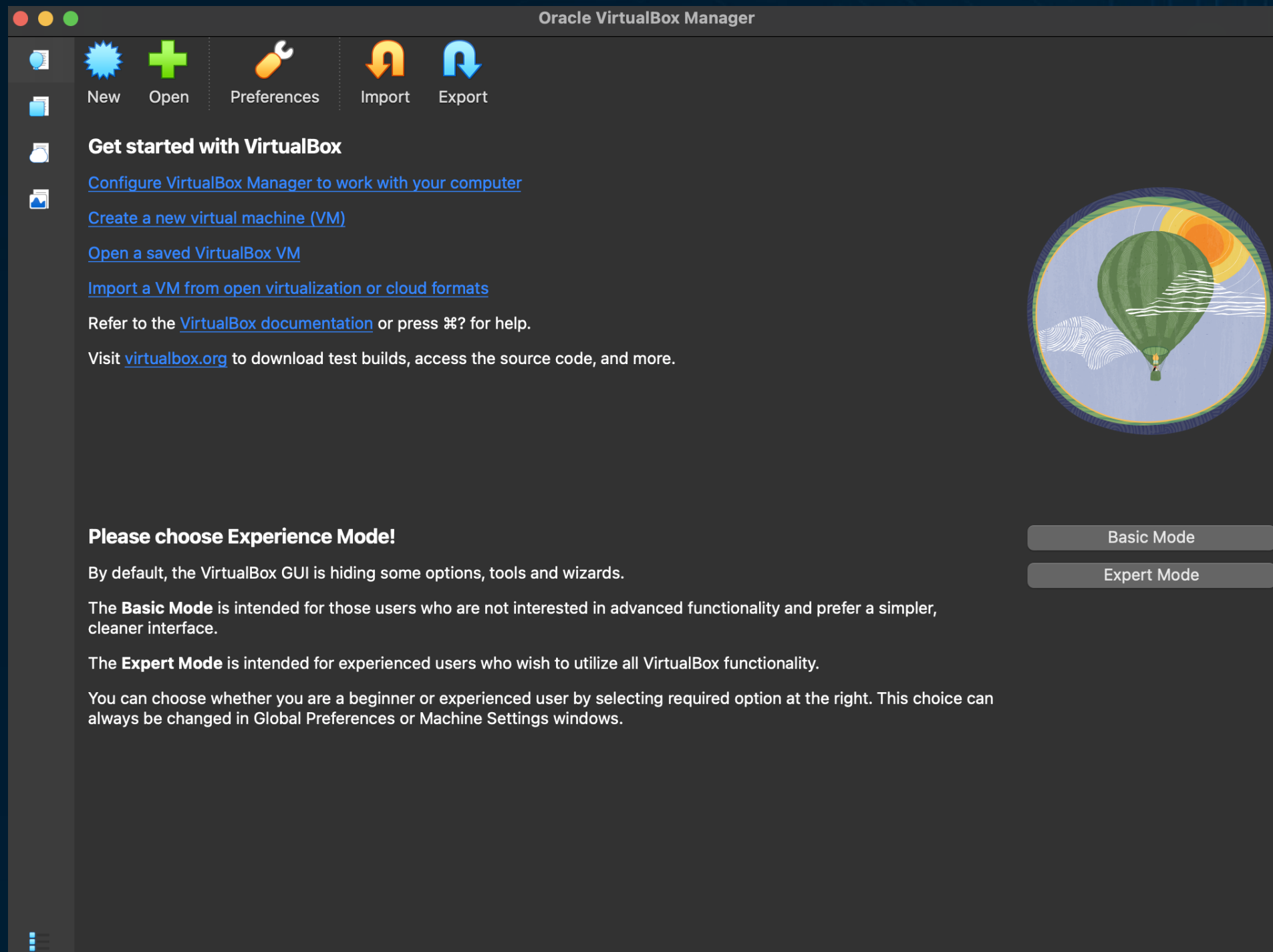
Installation Steps (Windows)

1. Right-click installer → Run as Administrator
2. Click Next through welcome screen
3. Keep default installation location: `C:\Program Files\Oracle\VirtualBox\`
4. Keep all features selected (default)
5. Warning: Network will briefly disconnect → Click Yes
6. Click Install
7. Allow driver installation when prompted
8. Click Finish

Installation Steps (Windows)



VirtualBox after installation



Lab 2: Configure VirtualBox First-Time Setup

Configure Default Machine Folder

1. Open VirtualBox
2. Go to File → Preferences (Ctrl+G)
3. Select General tab
4. Under Default Machine Folder click folder icon
5. Choose location with plenty of space (recommend dedicated folder)
 - Example: `D:\VirtualMachines` or `~/VirtualMachines`
 - Avoid C: drive if space limited
6. Click OK

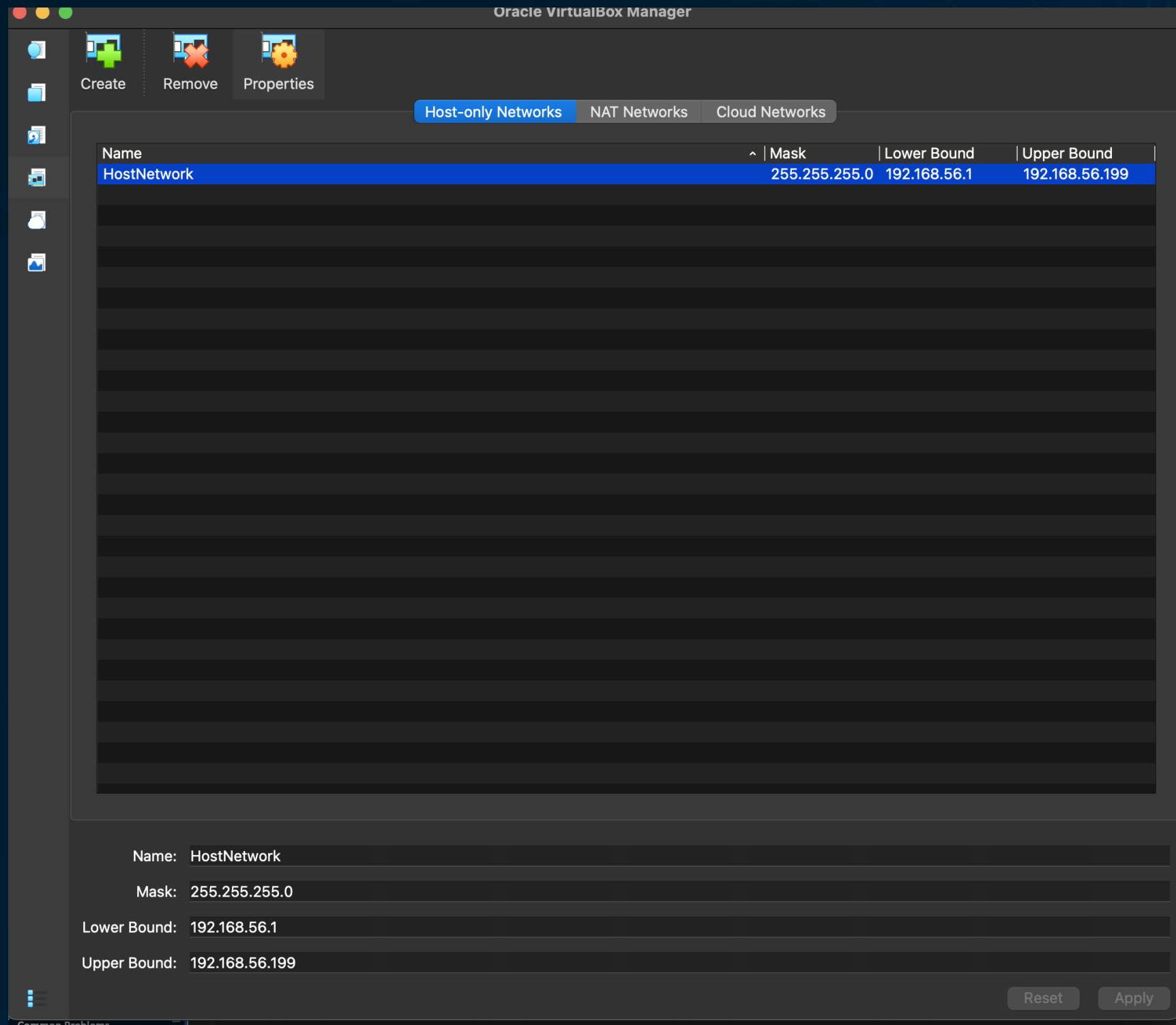
Configure Default Network Settings

1. In Preferences, go to Network tab
2. Select NAT Networks tab
3. Click + icon to add new NAT Network
4. Name it: `LabNetwork`
5. Network CIDR: `10.0.2.0/24` (default is fine)
6. Keep Supports DHCP checked
7. Click OK

Set Up Host-Only Network (for future labs)

1. Still in Network tab, select Host-only Networks
2. Click + icon to create
3. Name will be auto-generated (e.g., `vboxnet0`)
4. Click wrench icon to configure
5. Adapter tab:
 - IPv4 Address: `192.168.56.1`
 - IPv4 Network Mask: `255.255.255.0`
6. DHCP Server tab:
 - Check Enable Server
 - Server Address: `192.168.56.100`
 - Server Mask: `255.255.255.0`
 - Lower Address Bound: `192.168.56.101`
 - Upper Address Bound: `192.168.56.254`
7. Click OK

Set Up Host-Only Network (for future labs)



כל הזכויות שמורות לסייברלאבס טכנולוגיות בע"מ

Configure General Preferences

1. Input tab:

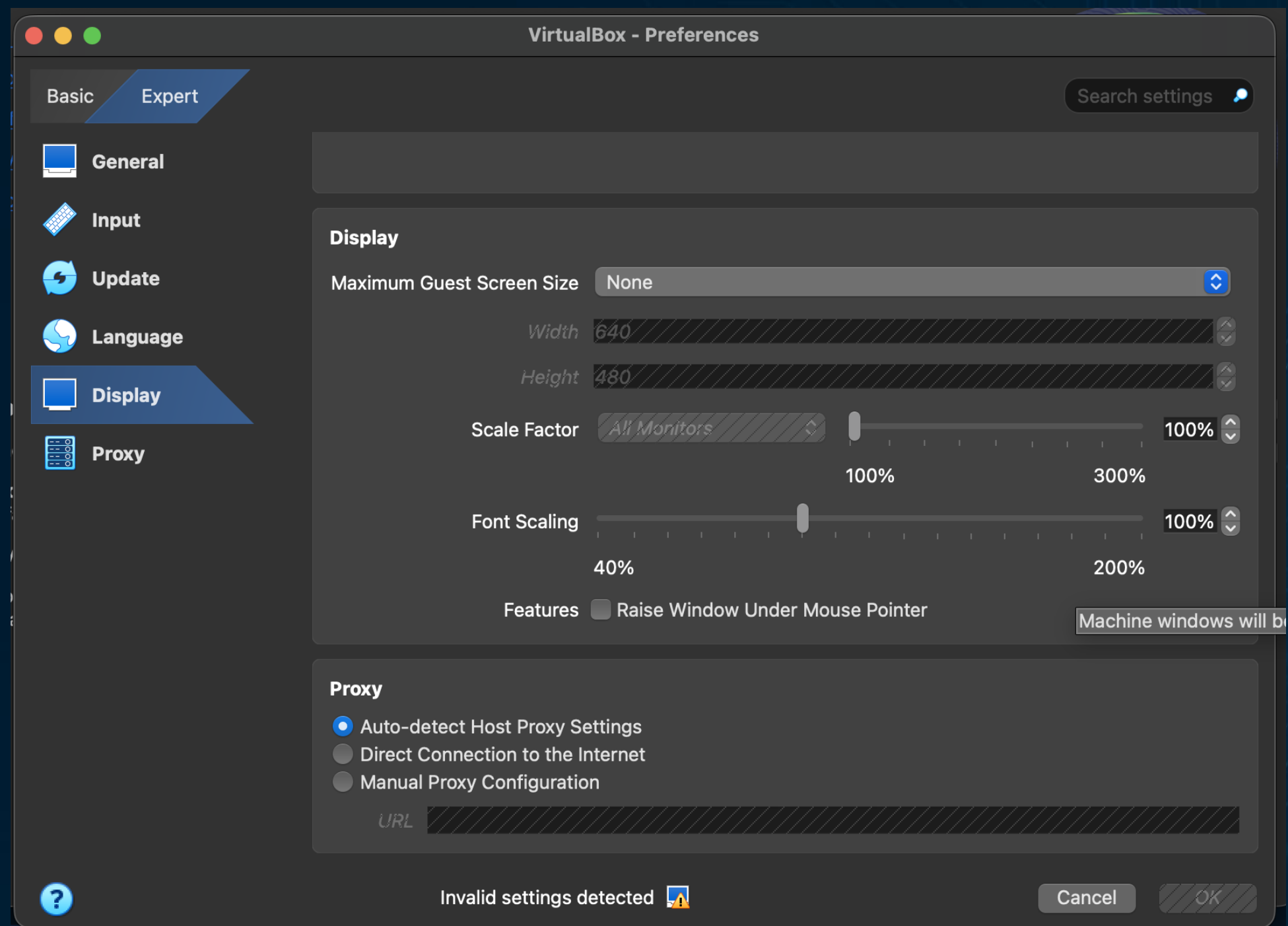
- Note the Host Key (usually Right Ctrl)
- This releases mouse from VM

2. Display tab:

- Set Maximum Guest Screen Size: None (or match your monitor)

3. Click OK to save all settings

Display Settings



Lab 3: Download & Configure Kali Linux

Step 1: Download Kali Linux VM (AMD64)

1. Open browser
2. Navigate to: ``https://www.kali.org/get-kali/#kali-virtual-machines``
3. Download VirtualBox version (64-bit)
 - File: ``kali-linux-YYYY.X-virtualbox-amd64.7z``
 - Size: approximately 3-4GB
4. Wait for download to complete

Download Kali Linux VM (AMD64)



www.kali.org/get-kali/#kali-virtual-machines

Installer Pre-built VMs ARM Mobile Cloud Containers Live WSL

Pre-built Virtual Machines

Kali Linux **VMware** & **VirtualBox** images are available for users who prefer, or whose specific needs require a virtual machine installation.

These images have the **default credentials** "**kali/kali**".

[Virtual Machines Documentation >](#)



VMware

3.4G torrent docs sum



VirtualBox

3.4G torrent docs sum



Hyper-V

3.4G torrent docs sum



QEMU

3.4G torrent docs sum

Step 2: Extract & Install Downloaded File

1. Download 7-Zip if not installed: `<https://www.7-zip.org>`
2. Right-click downloaded `.7z` file
3. Select 7-Zip → Extract Here
4. Wait for extraction (creates `.vbox` and `.vdi` files)
5. Note extraction location. Enter system password if prompted
- 6 .Open VirtualBox
7. Click Machine → Add
8. Navigate to extracted folder
9. Select `.vbox` file
10. Click Open

Configure VM Settings Part 1

1. Right-click Kali Linux VM → Settings/System Settings
2. Go to System → Motherboard tab
 - Base Memory: Allocate 4096 MB (4GB) if you have 16GB host RAM
 - Or 2048 MB (2GB) if you have 8GB host RAM
 - Boot Order: Check Hard Disk, Optical (uncheck Floppy)
3. Processor tab:
 - CPUs: Allocate 2 CPUs
 - Execution Cap: 100%

Configure VM Settings Part 2

Display Settings

4. Go to Display → Screen tab

- Video Memory: 128 MB
- Graphics Controller: VMSVGA
- Enable 3D Acceleration: Check

**Network Settings (NAT):

5. Go to Network → Adapter 1 tab

- Enable Network Adapter: Checked
- Attached to: NAT
- Adapter Type: Intel PRO/1000 MT Desktop
- Promiscuous Mode: Deny
- Cable Connected: Checked

6. Click OK to save all settings

Start & Login Kali Linux

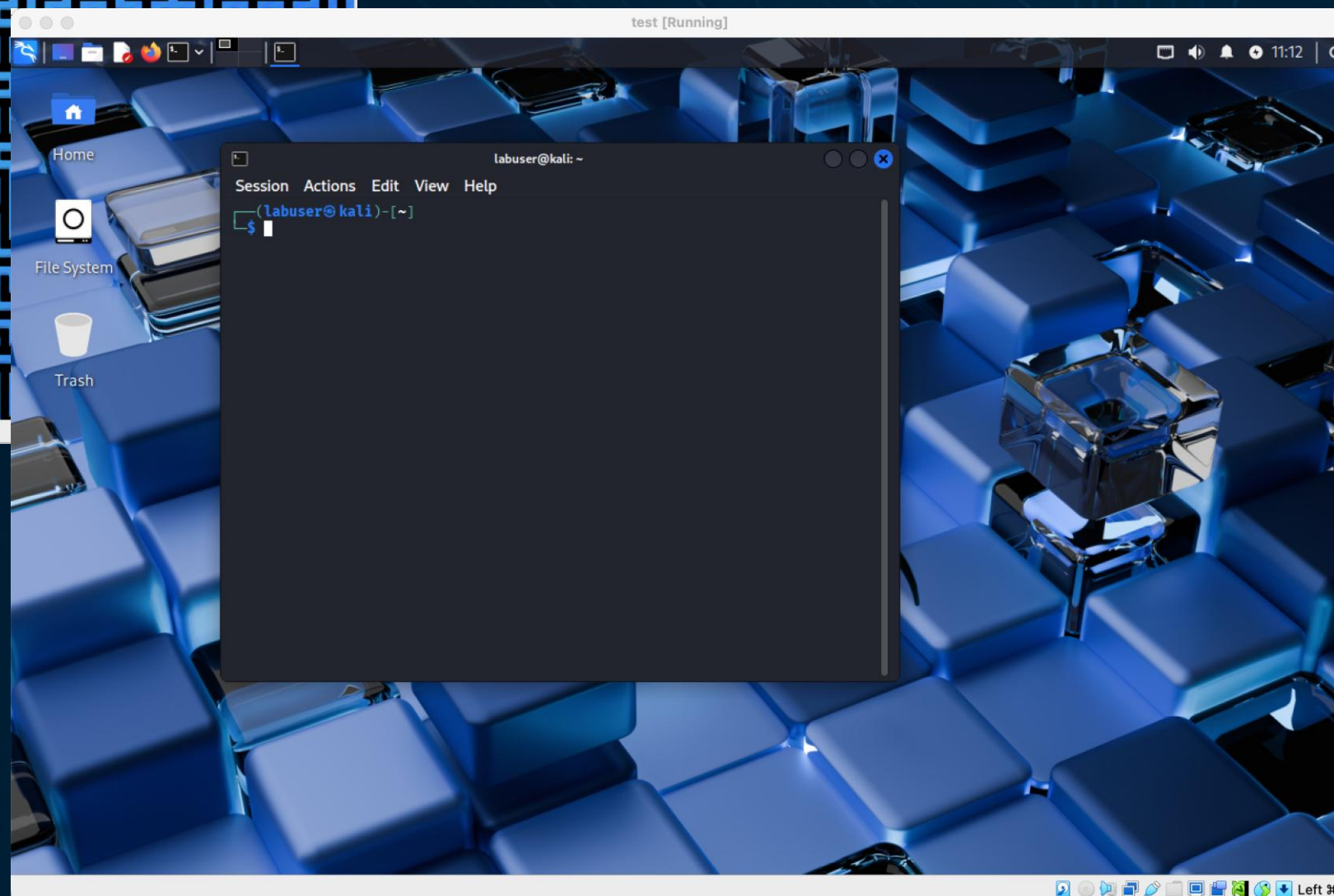
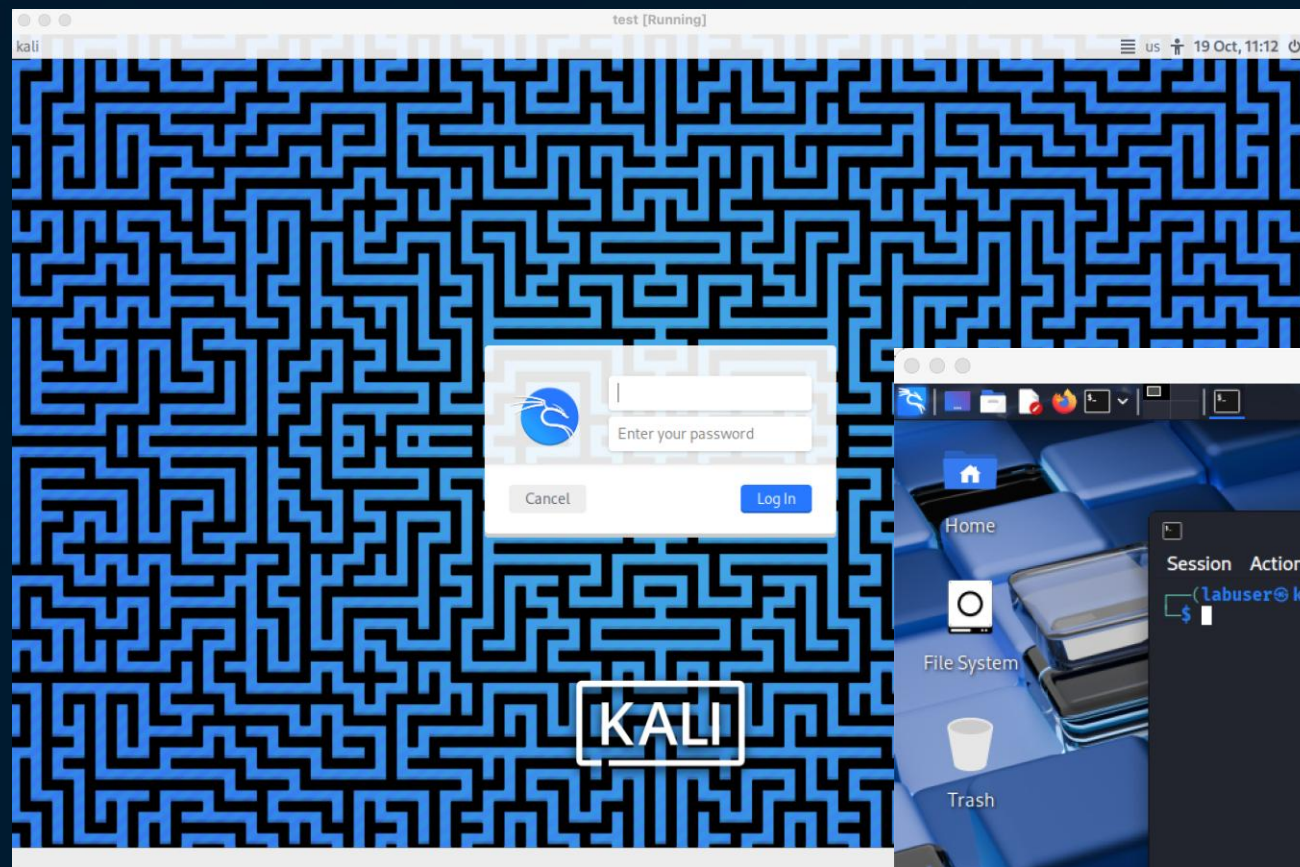
1. Select Kali Linux in VirtualBox
2. Click Start (green arrow)
3. VM window opens and Kali boots
4. Wait for login screen (30-60 seconds)

Default Credentials:

- Username: `kali`
- Password: `kali`

1. Enter credentials
2. Desktop loads

Start & Login Kali Linux



Confirm Network

1. Once logged in, open Terminal

- Click terminal icon in taskbar

2. Test internet connectivity:

```
ping -c 4 google.com
```

3. Check IP address:

```
ip addr show
```

Expected: IP address starting with `10.0.2.x` (NAT range)

4. Check DNS resolution:

```
nslookup google.com
```


Confirm Network

```
labuser@kali: ~  
Session Actions Edit View Help  
(labuser@kali)-[~]  
$ ping -c 4 google.com  
PING google.com (192.0.0.88) 56(84) bytes of data.  
64 bytes from 192.0.0.88: icmp_seq=1 ttl=255 time=206 ms  
64 bytes from 192.0.0.88: icmp_seq=2 ttl=255 time=207 ms  
64 bytes from 192.0.0.88: icmp_seq=3 ttl=255 time=208 ms  
64 bytes from 192.0.0.88: icmp_seq=4 ttl=255 time=207 ms  
  
— google.com ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3006ms  
rtt min/avg/max/mdev = 206.101/206.801/207.704/0.580 ms  
  
(labuser@kali)-[~]  
$ ip addr show  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:c8:21:b6 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86225sec preferred_lft 86225sec  
    inet6 fd17:625c:f037:2:74d8:c878:c3bc:e67d/64 scope global temporary dyna  
mic  
        valid_lft 86226sec preferred_lft 14226sec  
    inet6 fd17:625c:f037:2:a00:27ff:fec8:21b6/64 scope global dynamic mngtmpa  
ddr noprefixroute  
        valid_lft 86226sec preferred_lft 14226sec  
    inet6 fe80::a00:27ff:fec8:21b6/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(labuser@kali)-[~]  
$ nslookup google.com  
Server:      100.64.0.2  
Address:     100.64.0.2#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 192.0.0.88
```

Step 1: Pick Snapshot type

Option A: Create snapshot while running (live snapshot)

1. Keep Kali VM running
2. Clean up desktop if needed

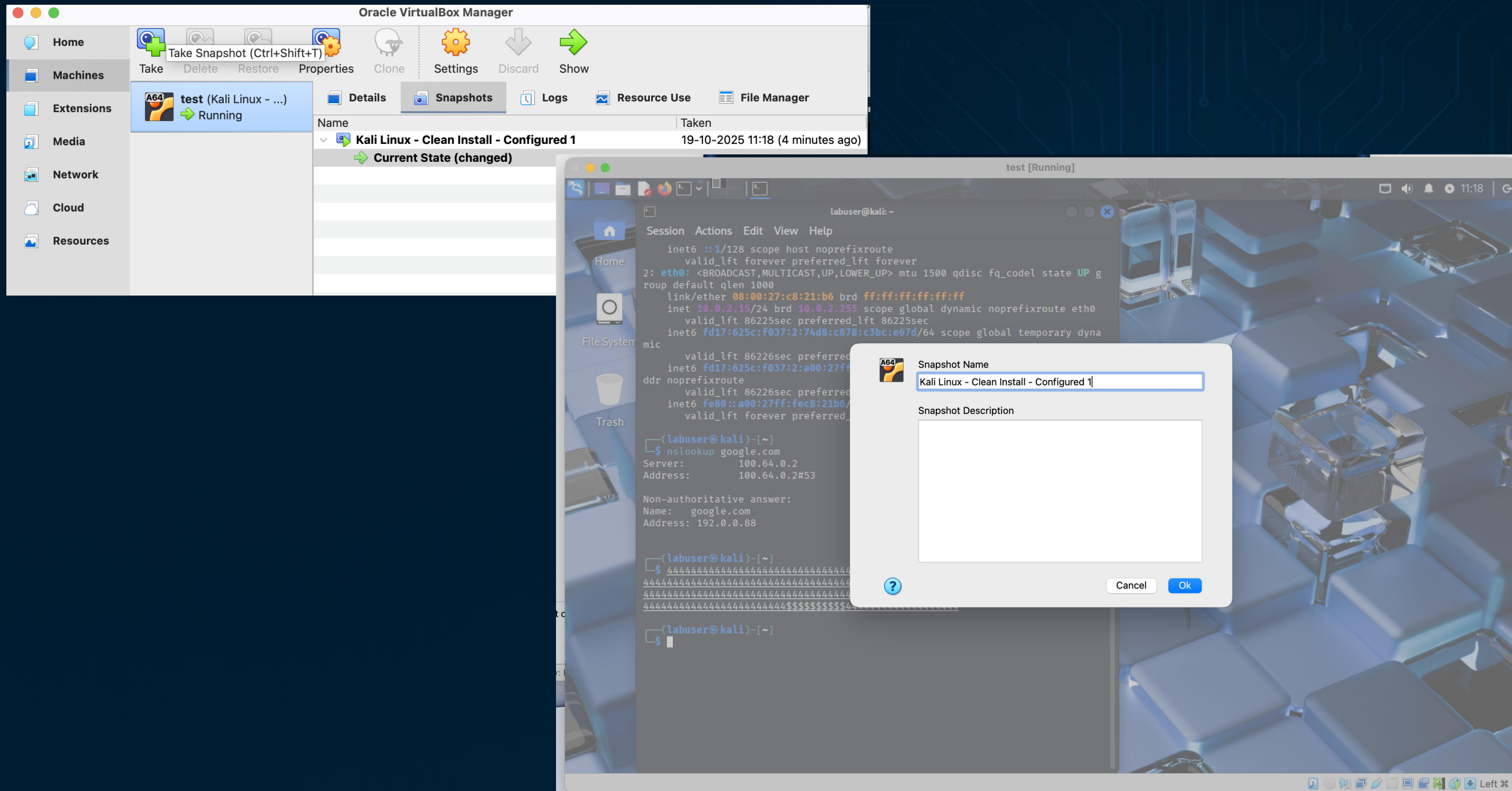
Option B: Create snapshot from powered off state (recommended)

1. In Kali, click Power icon → Shut Down
2. Wait for VM to fully power off
3. VM window closes

Step 2: Create Snapshot

1. In VirtualBox main window, select Kali Linux VM
2. Click Machine → Tools → Snapshots (or press Ctrl+Shift+S)
3. Or click the ≡ icon next to VM name → Snapshots
4. Click Take button (camera icon)
5. Follow Dialog

Step 2: Create snapshot

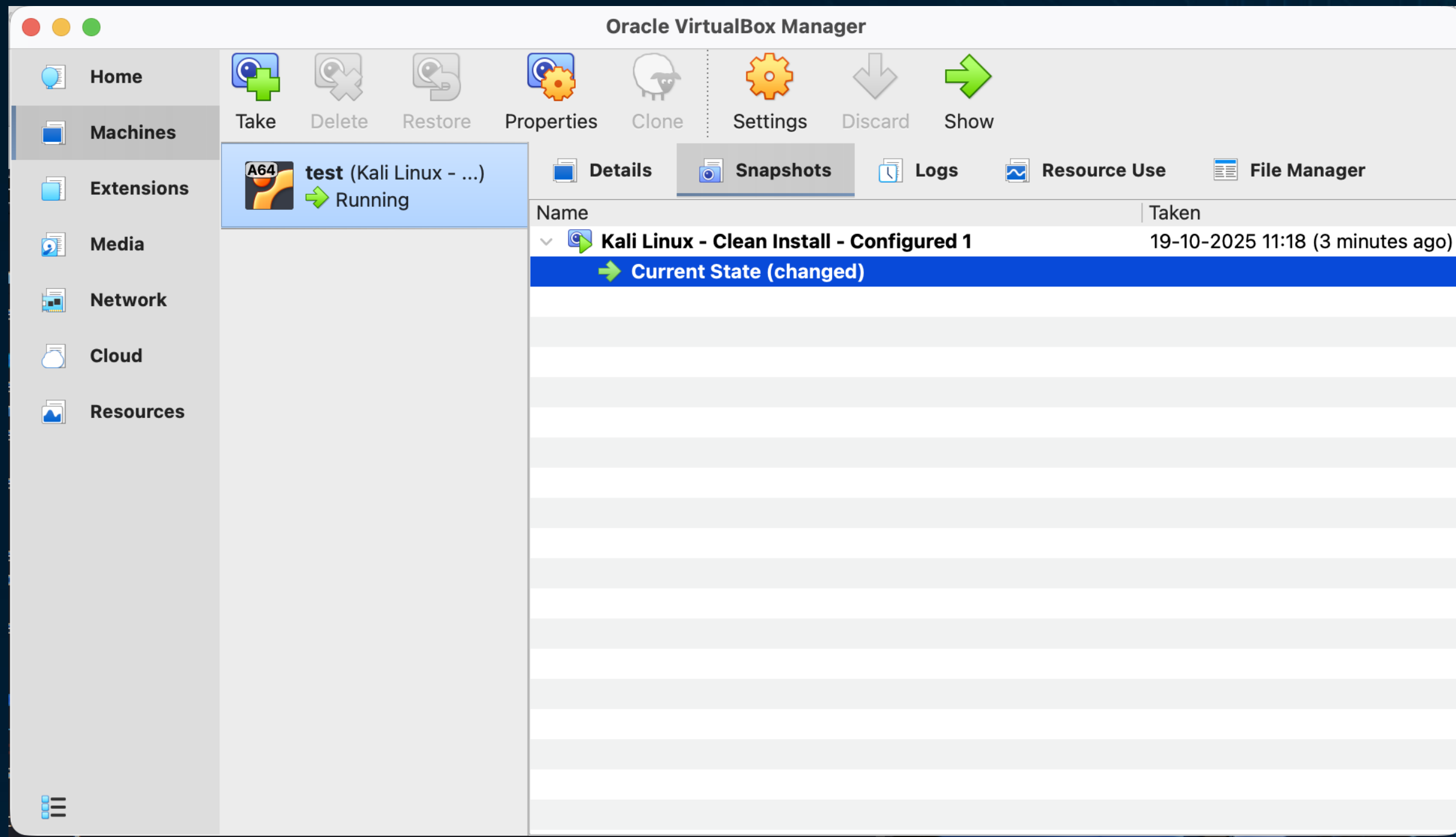


The screenshot displays the Oracle VM VirtualBox Manager interface. On the left, the 'Machines' list shows a VM named 'test (Kali Linux - ...)' in a 'Running' state. The main pane shows the 'Snapshots' tab for this VM, listing a single snapshot: 'Kali Linux - Clean Install - Configured 1', taken on '19-10-2025 11:18 (4 minutes ago)'. A 'Take Snapshot (Ctrl+Shift+T)' dialog box is open, prompting for a 'Snapshot Name' and a 'Snapshot Description'. The 'Snapshot Name' field is pre-filled with 'Kali Linux - Clean Install - Configured 1'. In the background, a terminal window within the VM shows network configuration details for 'eth0' and the output of an 'nslookup google.com' command.

Step 3: Restore Snapshot

1. Start Kali VM
2. Make a change (create file on desktop)
3. Shut down VM
4. Right-click snapshot → Restore
5. Confirm restoration
6. Start VM again
7. Verify: File you created is GONE (snapshot worked!)

Step 3: Restore Snapshot



Lab 5: Download and Install Windows 11 Trial VM

Download Windows 11 ISO

1. Download Windows 11 ISO from Microsoft

2. In VirtualBox, click New

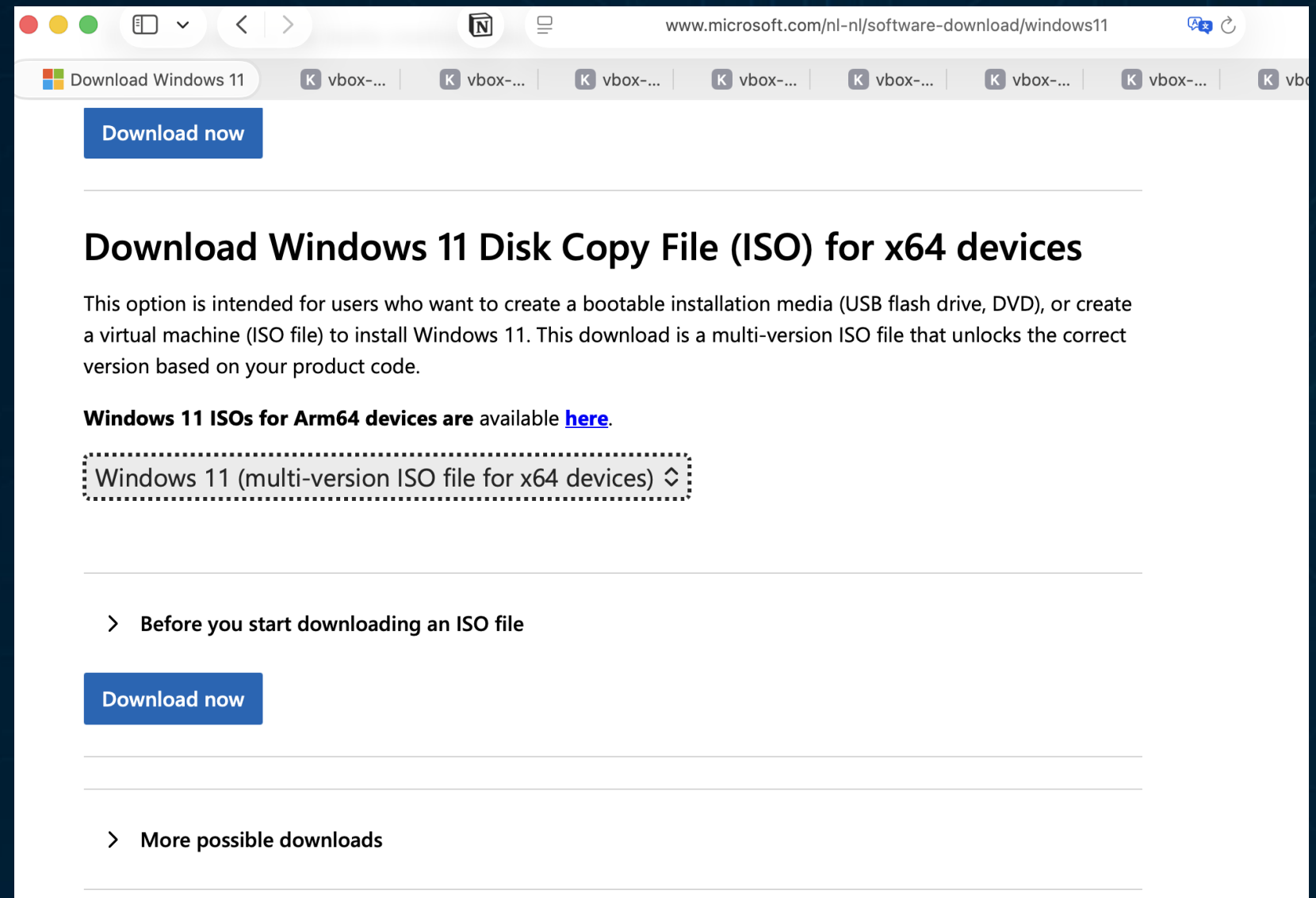
3. Configure:

- Name: `Windows 11 Lab`
- Type: Microsoft Windows
- Version: Windows 11 (64-bit)
- Memory: +4096 MB
- Create virtual hard disk now
- VDI format
- Dynamically allocated
- 80 GB size

4. Start VM

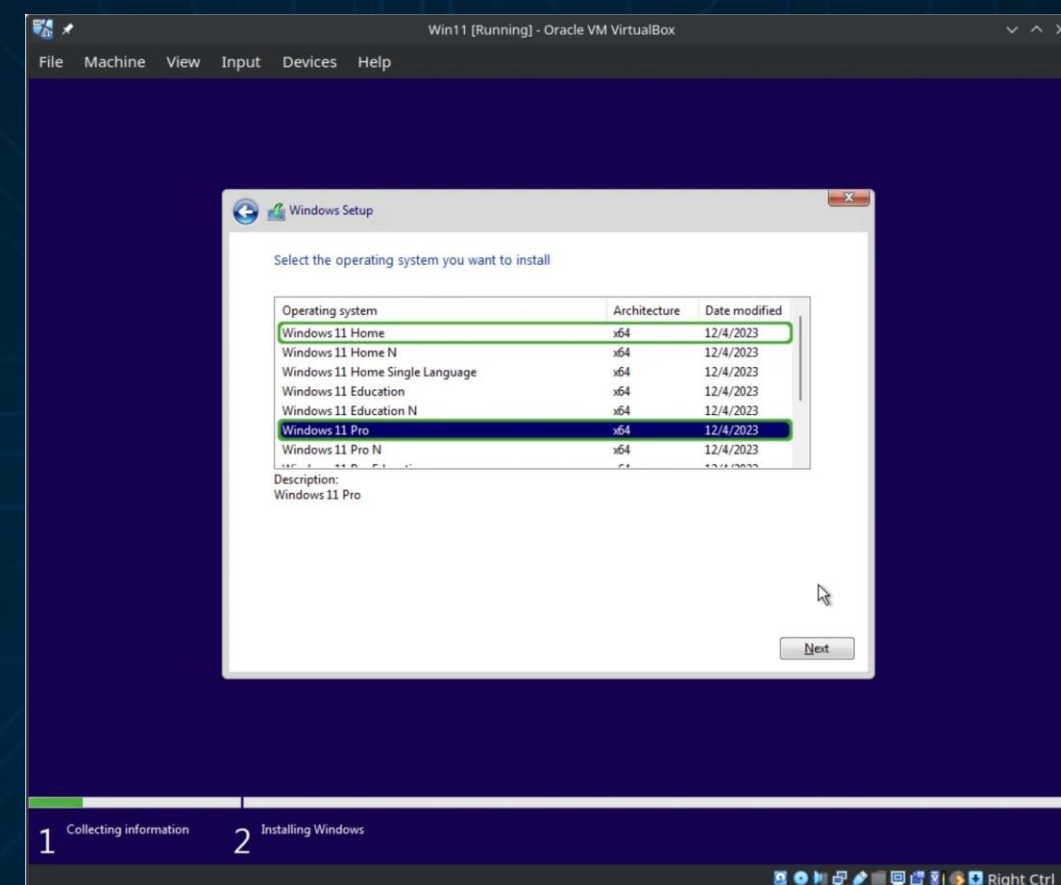
5. Select ISO file when prompted

6. Follow Windows installation wizard



Configure Windows 11

1. Select region/language
2. Choose Set up for personal use
3. Create local account (no Microsoft account needed for lab)
 - Username: `labuser`
 - Password: `Lab@2024!`
4. Disable telemetry options
5. Wait for desktop to load



Installing windows, virtual box settings



Create Virtual Machine

Summary

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.

✱ Machine Name and OS Type

Machine Name	Win11
Machine Folder	/srv/data/VM/VBox/Win11
ISO Image	/srv/data/ISO/Windows/Win11_23H2_English_x64v2.iso
Guest OS Type	Windows 11 (64-bit)
Skip Unattended Install	true

■ Hardware

Base Memory	8192
Processor(s)	4
EFI Enable	true

📀 Disk

Disk Size	80.00 GB
-----------	----------

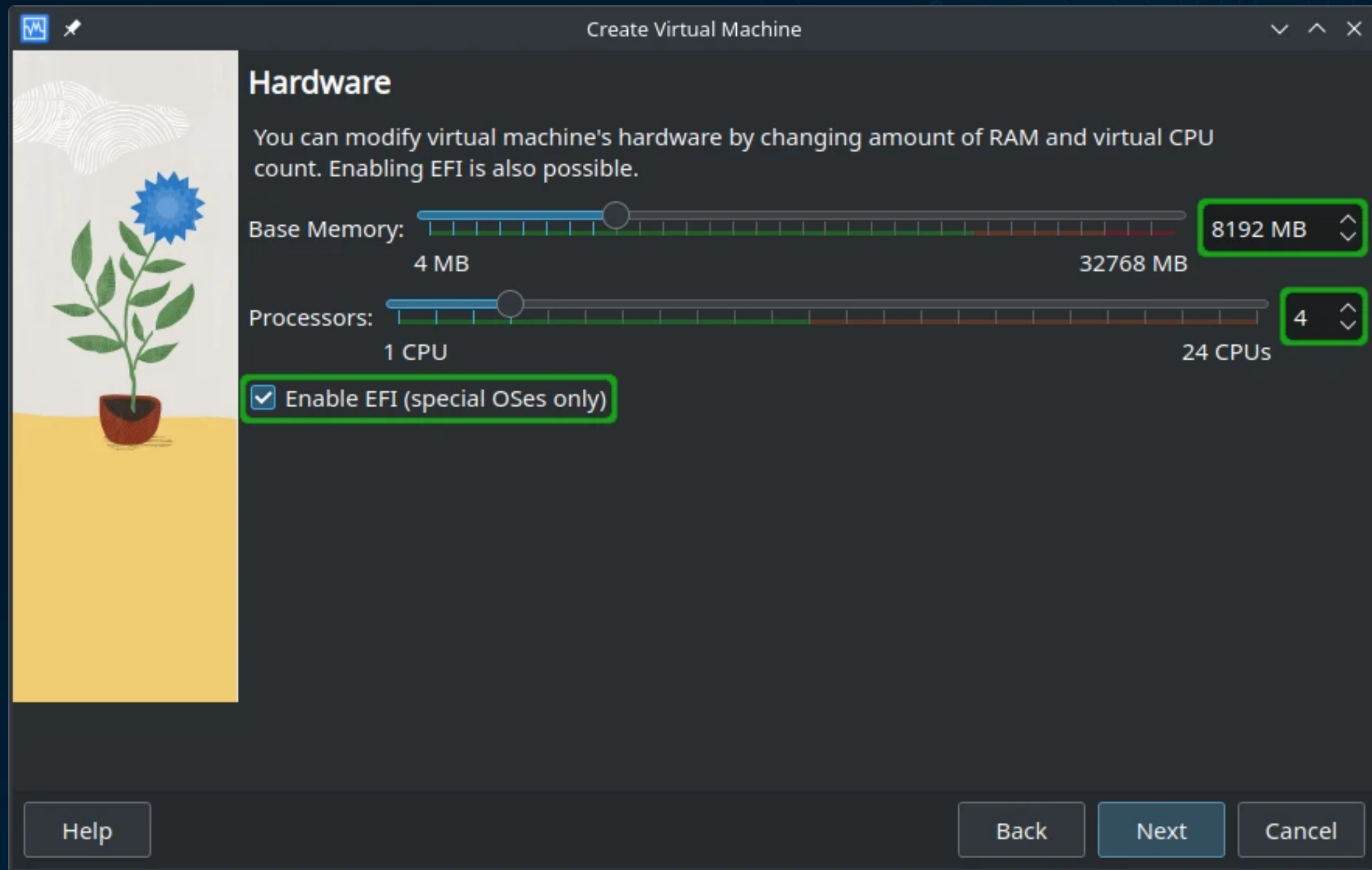
Help

Back

Finish

Cancel

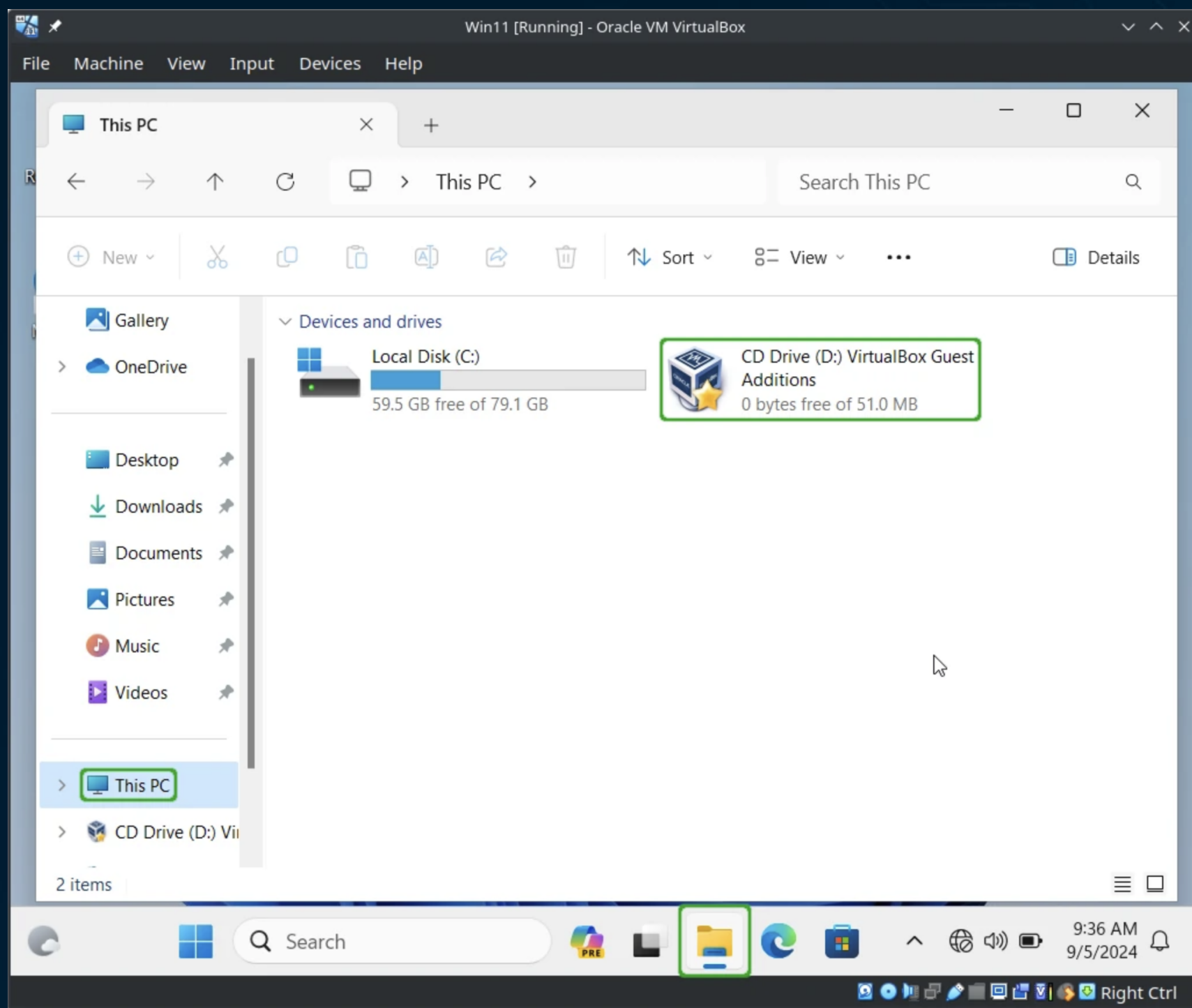
Installing Windows, virtual box settings



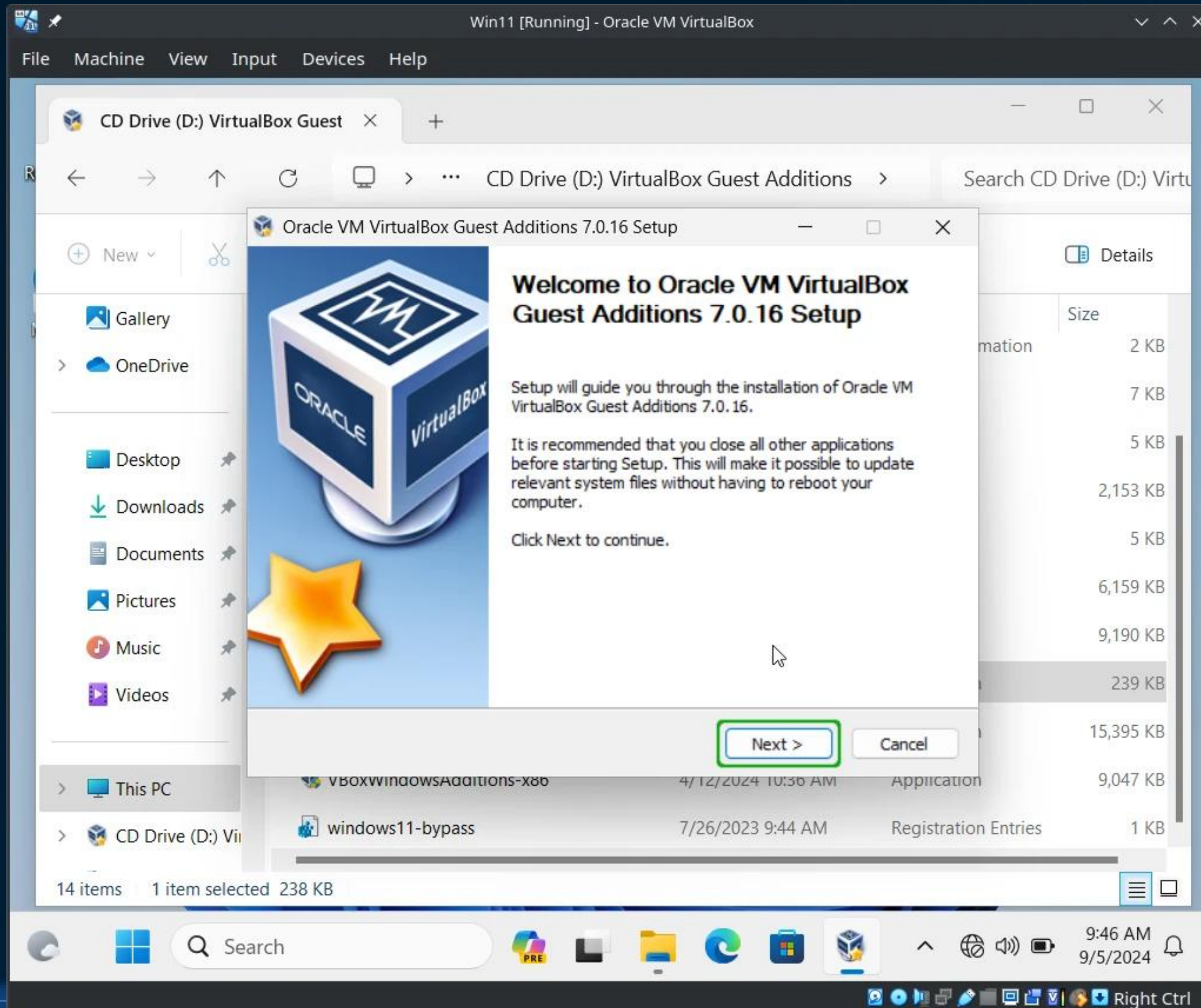
Critical: Improve performance 1

1. In VM window menu: Devices → Insert Guest Additions CD Image
2. Windows may auto-run installer
3. If not, open File Explorer → This PC → CD Drive (D:)
4. Double-click
`VBoxWindowsAdditions.exe`
5. Click Next through installer
6. Choose Reboot later when finished

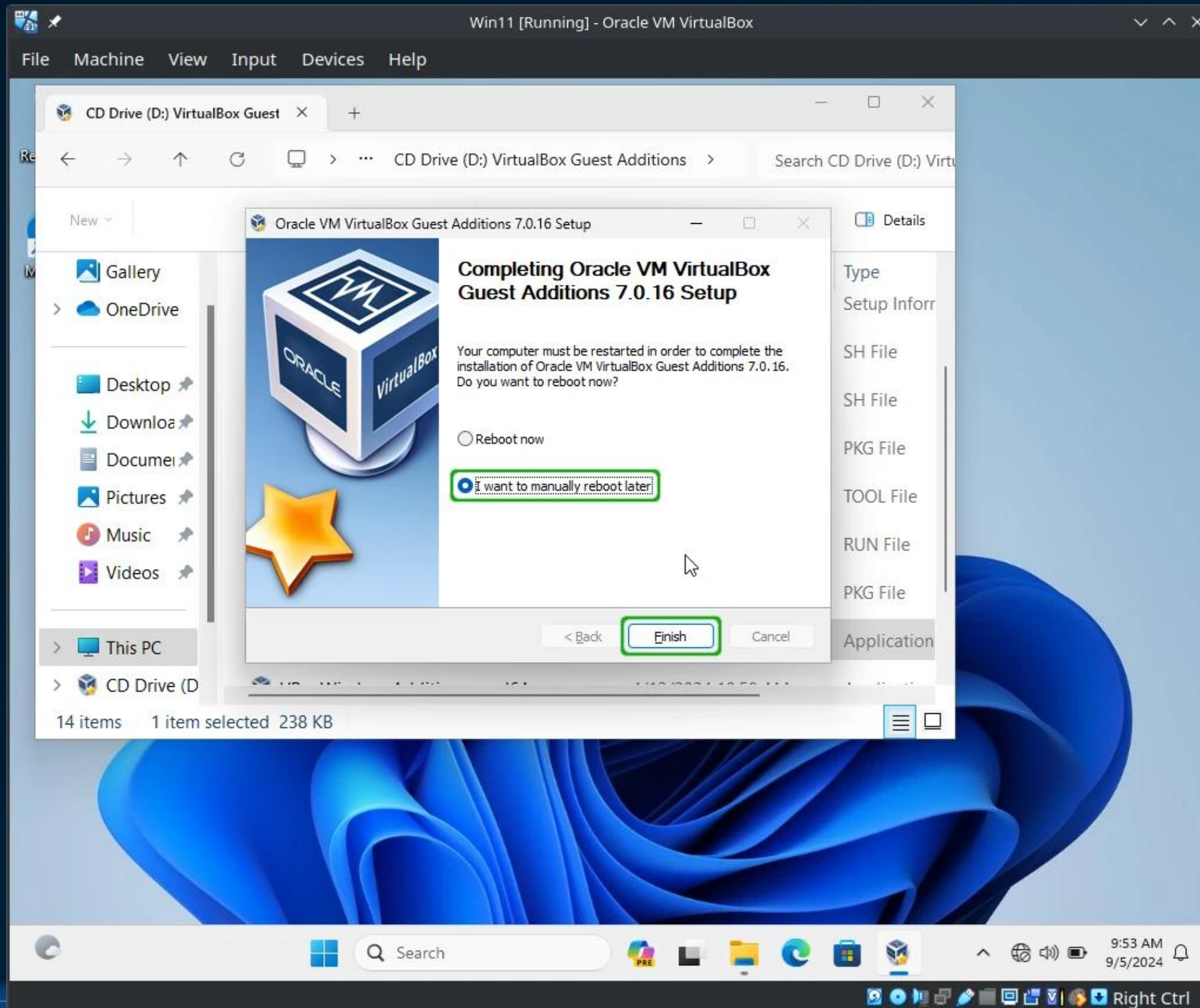
Critical: Improve performance 1



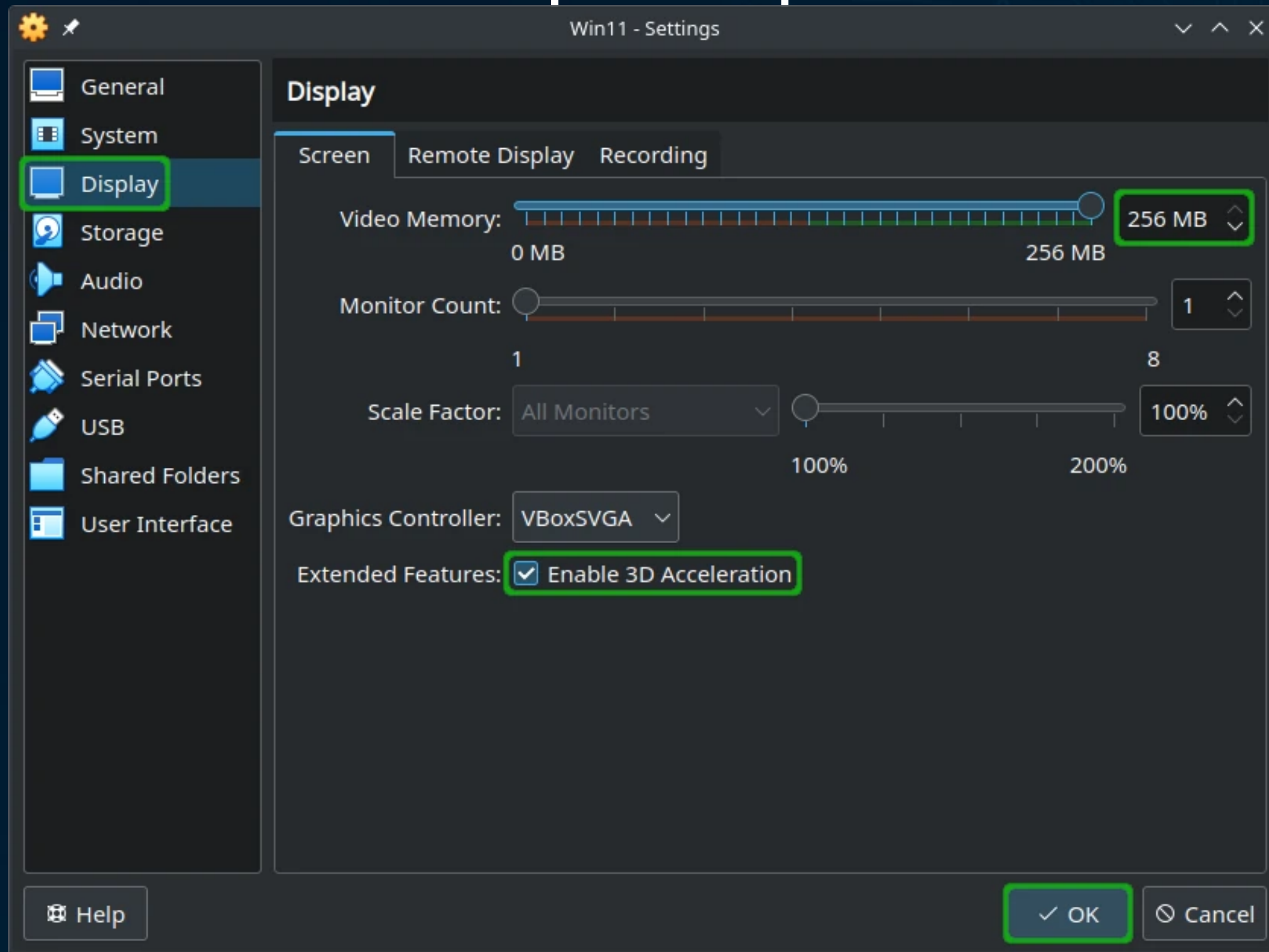
Critical: Improve performance 2



Critical: Improve performance 3



Critical: Improve performance 4



Configure windows 2

1. Disable Windows Defender (for labs only!):

- Open Windows Security
- Go to Virus & threat protection
- Manage settings → Turn OFF Real-time protection
- Turn OFF Cloud-delivered protection

2. Set static name

- Right-click Start → System
- Click Rename this PC
- Name: `WIN11-LAB`
- Restart when prompted

Verify Network Windows

. Open: Command Prompt

- Press `Win+R`
- Type `cmd` → Enter

2. Test connectivity:

`ping google.com -n 4`

3. Check IP configuration:

`ipconfig`

Expected: IP address starting with `10.0.2.x` (NAT range)

4. Check internet access:

- Open Microsoft Edge
- Navigate to any website

Create Snapshot Windows 11

1. In VirtualBox, select Windows 11 VM
2. Go to Snapshots view
3. Click Take button (camera icon)
4. Configure snapshot:
 - Name `Clean Install - NAT - Guest Additions`
5. Verify snapshot was created.
6. Restore snapshot like the Kali Snapshot

Lab 5B: Install Windows Server

Prerequisites

Required Before Starting

- VirtualBox installed
- NAT & Host-only networks configured
- Minimum 8 GB RAM (16 GB recommended)
- ISO: [Download Windows Server 2019 Evaluation](<https://www.microsoft.com/en-us/evalcenter/evaluate-windows-server-2019>)

Download Windows Server

← → ↻ info.microsoft.com/ww-landing-windows-server-2019.html ☆ 🔍 📁 | L Wor

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Microsoft Azure

Evaluate Windows Server 2019

Evaluate Windows Server 2019

Windows Server 2019 is the operating system that bridges on-premises environments with Azure services enabling hybrid scenarios maximizing existing investments.

- Increase security and reduce business risk with multiple layers of protection built into the operating system.
- Evolve your datacenter infrastructure to achieve greater efficiency and scale with Hyper-converged Infrastructure.
- Windows Server 2019 also enables you to create cloud native and modernize traditional apps using containers and micro-services.

Learn more about the features of [Windows Server 2019](#).

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Complete the form below.

* First name

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* Email

* Company name

* Country/Region

Install Server 1

1. Open VirtualBox → Click New
2. Configure:
 - Name: `WIN2019-DC`
 - Type: Microsoft Windows
 - Version: Windows 2019 (64-bit)
 - Memory: 4096 MB (4 GB)
 - Virtual Disk: 80 GB, VDI, dynamically allocated
3. Mount ISO when prompted
4. Start VM

Install Server 2

1. Language: English (US)
2. Edition: Windows Server 2019 Standard (Desktop Experience)
3. Custom Installation → New Partition → Next
4. Wait ~10–15 min
5. Create Local Admin:
 - Username: `Administrator`
 - Password: `P@ssw0rd!`

Snapshot 1: DC Clean install - unconfigured

Install Server 2

Windows Setup

Windows Server® 2019

Language to install: **English (United States)**

Time and currency format: **English (United States)**

Keyboard or input method: **US**

Enter your language and other preferences and click "Next"

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Windows Setup

Select the operating system you want to install

Operating system	Architecture	Date modified
Windows Server 2019 Standard Evaluation	x64	9/7/2019
Windows Server 2019 Standard Evaluation (Desktop Experien...	x64	9/7/2019
Windows Server 2019 Datacenter Evaluation	x64	9/7/2019
Windows Server 2019 Datacenter Evaluation (Desktop Experi...	x64	9/7/2019

Description:
This option installs the full Windows graphical environment, consuming extra drive space. It can be useful if you want to use the Windows desktop or have an app that requires it.

Next

Windows Server config 1

1. Rename computer → `DC01`

- Run: `sysdm.cpl` → Computer Name → Change

2. Set static IP:

Control Panel > Network Connections > Ethernet >
Properties

IPv4:

- IP: 192.168.56.10
- Mask: 255.255.255.0
- Gateway: 192.168.56.1
- DNS: 127.0.0.1

3. Disable Windows Firewall (for lab only)

כל הזכויות שמורות לסייברלאבס טכנולוגיות בע"מ

Promote to DC

1. Open Server Manager → Add Roles and Features
2. Choose:
 - Role-Based Installation
 - Check Active Directory Domain Services
 - Add Features → Next → Install
3. After install → Click yellow flag → “Promote this server to a domain controller”
4. New Forest:
 - Domain Name: `lab.local`
 - DSRM Password: `Dsrmp@ss!`

Snapshot Point 2— “AD + DNS Functional”

Verify Domain Controller

1. System reboots → Login as:

Administrator@lab.local

2. Open PowerShell → run:

Get-ADDomain

Get-Service -Name NTDS, DNS

3. DNS Server installed automatically

4. Verify FQDN resolution:

ping dc01.lab.local

nslookup lab.local

Create Organizational Unit & User

1. Open Active Directory Users and Computers
2. Create OU:
LabUsers
3. Add test user:
 - Username: `student01`
 - Password: `Lab@2024!`
 - Uncheck “User must change password”
4. Add to `Domain Users` group

Final Snapshot & Lab Reference

Take Final Snapshot:

> Name: `WIN2019-DC - Clean Install + AD Config`

Reference Summary:

VM Name, Role, Domain, IPs, Other remarks

Troubleshooting Windows Server

Common Issues

- Cannot promote to DC:
 - Check DNS client points to 127.0.0.1 |
- Network errors:
 - Verify Host-only Adapter bound to vboxnet0
- Slow boot:
 - Disable Defender real-time protection:
- AD tools missing:
 - Add via “RSAT: Active Directory Tools” |

Lab 6: Create a lab document



CYBERLABS

MODULE 1 - LAB REFERENCE SHEET Content

VIRTUALBOX INSTALLATION

- Version: [Your version]
- Extension Pack: Installed
- Default VM Folder: [Your path]

KALI LINUX VM

- VM Name: Kali Linux
- Username: kali
- Password: kali
- Network: NAT (10.0.2.15)
- RAM: 4GB
- CPUs: 2
- Snapshot: Clean Install - NAT Network



CYBERLABS

MODULE 1 - LAB REFERENCE SHEET Content

Cont.

WINDOWS 11 VM

- VM Name: Windows 11 Lab
- Username: [Your username]
- Password: [Your password]
- Network: NAT (10.0.2.15)
- RAM: 4GB
- CPUs: 2
- Snapshot: Clean Install - NAT - Guest Additions
- Trial Expires: [Date]

WINDOWS Server

- VM Name:
- Username: [Your username]
- Password: [Your password]
- Network: -
- CPUs:
- Snapshot:
- Trial Expires: [Date]

MODULE 1 - LAB REFERENCE SHEET Content

Cont.

IMPORTANT SHORTCUTS

- Release mouse: Right Ctrl (Host Key)
- Full screen: Right Ctrl + F
- Snapshots: Ctrl + Shift + S
- Settings: Ctrl + S

TROUBLESHOOTING NOTES

[Add any issues you encountered and solutions]

Lab 7: Prepare for the next module

Before the next module

Before Module 2 (Networking):

- Keep both VMs in "Clean Install" state
- Both should use NAT networking
- Have snapshots ready to restore if needed

Thank you